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MILITARY STANDARD

TECHNICAL REPAIR STANDARDS (TRS); HULL, MECHANICAL, AND ELECTRICAL (HM&E), PREPARATION OF (METRIC)



AMSC No. N3467 AREA TMSS

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Technical Repair Standards (TRS); Hull, Mechanical, and Electrical (HM&E), Preparation of (Metric)

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. A TRS is a technical document that covers the minimum requirements and procedures for the overhaul of an item to class B condition in accordance with OPNAVINST 4700.7F. The TRS is a standard, not simply a procedure or a substitute for a shipboard maintenance manual. It is intended that TRSs developed in accordance with the provisions contained herein shall provide specifying activities, user activities, and planning and estimating managers with an overhaul procedure plus the specific planned examination and test acceptance criteria to which an item must perform or be critically examined prior to final acceptance of the completed overhaul. User activities shall promulgate the TRS criteria in local overhaul implementing specifications for users to accomplish the work and evaluate its quality upon completion.

2. This standard covers the requirements and instructions for the preparation of TRSs for use in performing class B overhauls to Naval Sea Systems Command (NAVSEA) HM&E equipment and components with the objective of obtaining satisfactory service from the item between planned overhauls. This standard contains specific integrated instructions for the preparation of:

(a) TRS outline/book plan.
(b) TRS manuscripts and preliminary manuals.
(c) TRS reproducible copy.
(d) Planned overhaul repair records.
(e) Examination, test, and repair action records.
(f) TRS quality assurance provisions and data.
(g) Changes and revisions to TRSs.
(h) Standard TRS covers and title pages.

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1. SCOPE

1.1 General. This standard establishes uniform quality control, technical content, format, and style criteria for hull, mechanical, and electrical (HM&E) technical repair standards (TRSs).

1.2 Applicability. This standard is applicable to TRSs prepared for new systems (see 3.15), equipment (see 3.7), component acquisition (see 3.4), and items (see 3.8) that are currently in service and operational. This standard also is applicable to the revision of existing TRSs where more than 20 percent of the original pages are affected.

1.3 Implementation. This standard is to be used in the preparation of TRSs for systems, equipment and components, and will provide specific planned examination, overhaul (see 3.10), assembly and disassembly procedures, and test criteria for the repair (see 3.12) or overhaul of an item to a class B (see 3.11) condition.

1.3.1 Objective. The objective of this standard is to provide instructions for the preparation of TRSs that stress journeyman-level workmanship quality standards in the accomplishment of an item overhaul and to provide final acceptance criteria.

1.3.2 Compliance. TRSs developed in accordance with the provisions of this document require overhaul and repair activities to comply with all critical examinations and acceptance tests (see 3.1) and criteria, to include dimensional criteria.

1.4 Technical repair standards. HM&E TRSs provide, on a systematic and comprehensive basis, the detailed instructions, procedures, and supporting illustrations for an effective and efficient method of performing a class B overhaul of the items covered in the document.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. Unless otherwise specified, the following specifications, standards, and handbook of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-9868 - Microfilming of Engineering Documents, 35mm; Requirements for.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content, Requirements for.
MIL-M-38748 - Microfiche; for Engineering/Technical Data, Reports, Studies, and Related Data, Requirements for.
MIL-M-38784 - Manuals, Technical: General Style and Format Requirements.

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MILITARY (Continued)

MIL-M-85337 - Manuals, Technical: Quality Assurance Program; Requirements for.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents.
DOD-STD-480 - Configuration Control - Engineering Changes, Deviations, and Waivers.
DOD-STD-1685 - Comprehensibility Standards for Technical Manuals. (Metric)

HANDBOOK

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

2.1.2 Other Government publications. The following other Government publications form a part of this standard to the extent specified herein.

PUBLICATIONS

DEPARTMENT OF DEFENSE (DoD)

DoD 5220.22-M - Industrial Security Manual for Safeguarding Classified Information, DD Form 254.

OFFICE OF THE CHIEF OF NAVAL OPERATIONS INSTRUCTION (OPNAVINST)

OPNAVINST 5100.23 Series - Navy Occupational Safety and Health (NAVOSH) Program Manual.

UNITED STATES GOVERNMENT PRINTING OFFICE (GPO)

Style Manual - Library of Congress Catalog No. Z253.U58.

(Copies of specifications, standards, handbooks, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DoDISS.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Nongovernment standards are generally available for reference from libraries. They are also distributed among nongovernment standards bodies and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this specification shall take precedence.

3. DEFINITIONS

3.1 Acceptance tests. An acceptance test is an examination and test of an item after completion of its overhaul or repair to ensure that the item satisfies minimum specified performance requirements.

3.2 Adequacy. Adequacy is a depth of scope of coverage sufficient to support all tasks and functions at the prescribed level of the user, consistent with the equipment to be used and the mission environment in which the manual is to be used.

3.3 Assembly. An assembly is a number of parts or subassemblies, or any combination thereof, joined together to perform a specific function and capable of being disassembled.

3.4 Component. A component is a composite fabricated unit (generally complete within itself) that is designed to perform a stated service when installed in its proper position (for example, lube oil cooler, overspeed trip, strainer, and so forth).

3.5 Critical procedures. Critical procedures are those procedures which:

(a) are necessary to restore the equipment to performance specifications,
(b) if not accomplished, would prevent the equipment from operating between overhauls,
(c) contain safety-related or precautionary measures designed to minimize personal injury or equipment damage.

3.6 Critical readings. Critical readings are those readings which must be maintained within specified tolerances to prevent serious degradation in unit performance or shortened service life. Operation of the unit with critical readings out of tolerance also increases the possibility of damage to the unit or injury to personnel.

3.7 Equipment. Equipment is a component or components and necessary assemblies, subassemblies, and parts connected or associated together to perform an operational function (that is, boilers, feed pumps, motor generator sets, and so forth).

3.8 Item. An item is a nonspecific term used to denote any product, including systems, components, equipment materials, parts, subassemblies, sets, accessories, and so forth.

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3.9 Mandatory overhaul replacement parts. Mandatory overhaul replacement parts are those parts or components that must be replaced, regardless of condition, to restore a repairable item to a predetermined level of readiness.

3.10 Overhaul. Overhaul is the process of reconditioning a system or equipment to conform to the stated performance and technical specifications of the system or equipment with a life expectancy equivalent to similarly configured new systems or equipment. Overhaul is also performed to repair or replace parts and components that have failed or are of marginal quality due to wear, deterioration, or damage so as to preclude premature failure. Installation of authorized approved engineering or field changes may be included as part of the overhaul.

3.11 Class B overhaul. Class B overhaul is work that requires such overhaul or repairs as will restore the operating and performance characteristics of a system, subsystem, or component to its "original" design and technical specifications. If it is required to restore the operating and performance characteristics of an item to other than its original design and technical specifications, it must be so specified and the performance criteria defined. ShipAlts, OrdAlts, field changes, and modifications, even if applicable, are not to be accomplished unless specified by the customer. Where ShipAlts, OrdAlts, MachAlts, field changes and modifications have already been accomplished, they shall not be undone. Maintainance adjustment and calibration routines specified by the applicable instruction manual, unless superseded by authority, are required. The repair activity shall demonstrate that the end product successfully meets all performance criteria specified by the governing specifications.

3.12 Repair. Repair is work necessary to restore an unserviceable system or component to operational status without change in design, materials, number, location, or relationships of the component parts that have failed or are of marginal quality due to wear, deterioration, or damage. The repair process includes all necessary adjustment, alignment, and calibration procedures.

3.13 Reverse engineering. Reverse engineering is the process of generating from new or reconditioned equipment or subcomponents the technical data necessary to establish the minimum standards of acceptance required to accomplish a particular level of overhaul when this data is unavailable from existing documentation. (This process may include disassembly and measurement of an existing unit operating within design parameters.)

3.14 Subassembly. A subassembly is a portion of an assembly or unit which is replaceable as a whole, but also has a part or parts which are individually replaceable. (The distinction between an assembly and a subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another, where it forms a portion of an assembly.)

3.15 System. A system includes two or more equipments (sets) or components, each having its own identity and nomenclature, arranged and interconnected to perform a specific operation or function.

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3.16 Usability. Usability is the degree to which a user can apply the TRS in performing a specific task described by the TRS. Usability of a TRS by an appropriate user would be indicated by error-free performance of the described task on the first attempt. To be usable, a TRS must be operationally suitable; that is, it must be legible in the work environment involved and must not specify procedures, tools, or required skills uncharacteristic of the overhaul site in which it is to be used.

3.17 Validation. Validation is the process by which the TRS preparing activity tests a TRS for technical accuracy, adequacy, comprehensibility, and usability. It is conducted at the preparing activity's facility or at an overhaul site and involves performance of procedures, including checkout, calibration, removal, installation and alignment instructions, and associated checklists. Disassembly and reassembly, removal and installation procedures are validated by performance or simulation as approved in writing by the acquisition activity. Other data (such as part numbers, schematic diagrams, illustrations, and wiring data) are checked against current source data.

3.18 Verification. Verification is the process by which the acquisition activity tests a TRS to determine if it is accurate and adequate for the overhaul, repair, or restoration of the system, equipment, or component to which it applies. Verification may include preparing activity assistance and support. It is the final quality assurance iteration by the acquisition activity for acceptance of the TRS.

4. GENERAL REQUIREMENTS

4.1 Security classification. Unless otherwise specified herein, a TRS shall not contain classified information. When the TRS must contain classified data, the security classification shall be assigned by the acquisition activity. When classified information is included in a TRS, the marking, handling, and production of the classified material shall be in accordance with DoD 5220.22-M. On foldout pages of classified TRSs, the security classification shall be visible when the printed page is either folded or open.

4.2 Status reports. The preparing activity shall deliver periodic status reports. TRS status reports may be combined with other management level reports.

4.2.1 Format and style. Status reports may be typewritten on 8-1/2 x 11-inch commercial bond paper using the contractor format or in a format similar to figure 1.

4.3 TRS outline/book plan. A TRS outline/book plan shall be prepared prior to development of the TRS manuscript. The outline/book plan shall consist of a narrative outline and a book plan.

4.3.1 Narrative outline. A brief narrative in outline form shall accompany the book plan. The outline shall include the following.

(a) A brief overview of the overhaul strategy, accompanied by a repair process flow chart (see figure 1).
(b) A description of the problems regarding the interpretation and application of this standard.

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(c) A list of apparent conflicts between this standard and other guidance.
(d) A listing of all technical documentation (such as, technical manuals, drawings, including date and revision number, performance specifications, engineering change data, approved parts lists (APLs), and manufacturer's bulletins) to be used in the preparation of the TRS.
(e) Any recommendations for additional art work which may need to be developed.

4.3.2 Book plan. A book plan shall be developed based on the following:

(a) The TRS book plan shall be prepared in accordance with the requirements of this standard. Each paragraph title or notation listed in section 5 of this standard shall be followed by a brief narrative of the data to be presented.
(b) The range and depth of the TRS book plan shall be based on the maintenance engineering or logistic support analyses and shall include the following:

(1) The minimum procedures and testing requirements to ensure the class B overhaul is performed in an effective manner by a journey-level mechanic.
(2) A listing of the mandatory replacement parts consistent with the required period between overhauls and supporting historical data.
(3) A listing of all restricted parts that, due to their critical nature, must be acquired from a qualified source.
(4) A listing of contingency materials that may be needed to complete the overhaul (see 5.12.2).

(c) An illustration guide and a table guide shall be provided. Each illustration and table listed shall be described. The illustration plan shall contain figure numbers, title, information content, and approximate size and nature of the illustration (schematic, line drawing, exploded view, and so forth). A sample of each drawing type referenced shall be included.
(d) The estimated total number of text and illustration pages per chapter shall be listed.
(e) Data required for the front matter including a brief description of the material to be presented in the foreword shall be prepared (see 5.4).

4.3.3 Acceptance. The narrative outline and book plan shall be developed prior to development of the TRS. If unacceptable, the unacceptable portions shall be corrected and resubmitted until accepted. The accepted outline/book plan shall be available for review at all scheduled in-process reviews. Acceptance of the outline/book plan by the acquisition activity shall not waive any requirement within the scope of this standard or reference standard.

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4.3.4 Updating. The outline/book plan shall be kept up-to-date throughout the TRS development and until acceptance of the TRS manuscript. Improved data presentation shall be a prime justification in preparing changes to the outline and book plan.

4.4 Manuscript TRS. The manuscript shall contain all data required for the final TRS. It shall be technically accurate and adequate, and shall represent the configurations of the items for which the TRS is prepared. The manuscript shall be validated. Corrections required as a result of in-process reviews and validation shall be incorporated. Copies of the manuscript TRS shall be provided for technical evaluation as follows:

(a) Corrected/updated text and a complete final front matter shall be properly incorporated.
(b) Tables shall be corrected/updated and properly incorporated.
(c) Final size illustrations shall be corrected/updated and properly incorporated (see 4.13.7). Illustrations may be foldout pages but shall not exceed 279 millimeters (mm) (11 inches) in height and 114 centimeters (cm) (45 inches) in length with a maximum image area of 902 by 254 mm (35-1/2 by 10 inches).
(d) The word "MANUSCRIPT" shall be placed on the cover and title page in bold-faced, 18-point, capital letters, above the TMIN.
(e) A signed, dated validation certificate (see figure 2) shall accompany the review manuscript.
(f) Engineering judgment records (EJR) (see 4.7.1 and figure 3) shall accompany the review manuscript.
(g) Unclassified manuscripts shall be duplicated locally and loose leaf bound with screw fasteners in pressboard or equivalent binders or placed in a 3-ring notebook.

4.5 Preliminary TRS. When required the preliminary TRS shall be prepared (see appendix) in accordance with the content and format requirements of the manuscript copy and the following:

(a) The words "PRELIMINARY ISSUE" shall be placed on the cover and title page above the NAVSEA TMIN. Letters shall be capitalized and shall be 18-point bold-face type.
(b) The preliminary TRS shall be complete in all respects and shall reflect the manuscript with all comments acted upon. The preliminary TRS shall be submitted for final review and NAVSEA signature. Should additional changes be required, the developing activity shall revise accordingly and resubmit to NAVSEA for signature. At time of submittal of the preliminary TRS, the developing activity shall obtain a distribution list from the cognizant maintenance activity.
(c) The copy and integrally related artwork (including schematics, wiring diagrams, and block diagrams) shall be of sufficient darkness to reproduce clearly at required reproduction size without additional treatment. Illustrations shall be final size, complete with figure numbers and titles. Reproducible copy shall be as specified in 4.6 for final TRSs.

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4.6 Final TRS. The TRS reproducible copy is the property of the acquisition activity. The preparing activity shall deliver the approved TRS camera-ready reproducible copy as directed by the acquisition activity. The reproducible copy and integrally related artwork (including schematics, wiring diagrams, and block diagrams) shall be of sufficient darkness to reproduce clearly at required reproduction size without additional treatment. Illustrations shall be final size, complete with figure numbers and titles. The following shall apply:

(a) Running heads shall include the assigned NAVSEA TMIN (see 4.8) in the center and 13 mm (1/2 inch) from the top.
(b) Running feet of text pages shall include "Original" at the inboard edge and the page number at the outboard edge (see 4.8).
(c) There shall be no drawing data, other than horizontal lines, placed closer than 3 mm (1/8 inch) from the image area limit.
(d) Lines shall not be vertically displaced during the transition from one image area to the next.
(e) Minimum printed type size for text shall be 10-point elite type, 12 characters per inch. Nomenclature, callouts, tabular material, and symbols on illustrations shall be upper-case with 7-point (2.5 mm or 0.10 inch) minimum printed size.
(f) Text pages may be typed, and reproduced mechanically or by a computer typesetting system. The text shall be single column, justified left column format. Justified left and right margin format is approved when there is no additional cost to the Government.
(g) Spacing and lettering shall conserve space without reducing the usability or clarity of material. Letters, lines, and symbols shall be of a uniform contrast throughout the text material and shall not touch.
(h) Text pages shall be formatted to fall within a single 203 by 267 mm (8 by 10-1/2 inch) page when reduced.
(i) When reproduced, artwork shall be clear, adequate, and legible. Line drawings and illustrations rather than photographs shall be used.
(j) The line density for graphic material and background area shall be uniform to achieve the highest contrast for readability and reproducibility. Parallel lines on schematic diagrams shall be not less than 1.6 mm (1/16 inch) apart when reduced to printed size.
(k) The density of information presented shall be such that the spacing between lines of characters shall be not less than one-half the character height used.
(l) Corrections required, due to verification and review of the manuscript/preliminary TRSs, shall be incorporated. Copy shall be provided for each page of the TRS.
(m) Page elements not imprinted on the page, including corrections, shall be fastened to the page in such a manner to permit repeated handling of the reproducible copy over a period of years without the possibility of losing stripped-in portions.

4.7 Technical changes and deviations. Justification for changes to existing processes, clearances, tolerances, materials, or the development of new procedures must be submitted with the manuscript or preliminary TRS.

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4.7.1 First time acquisitions. Concurrent with the first acquisition of a new system, equipment or component designs, a concise, formal engineering judgment record (EJR) shall be prepared for each data entry (see appendix). The format of a typical EJR is shown on figure 3.

4.7.2 Reacquisitions. For TRSs or TRS change pages prepared for existing equipment or systems a formal EJR is not required for each data entry. Where the preparing activity proposes changes or deviations from existing documentation with regard to clearances, tolerances, materials, procedures, or the implementation of new procedures, worksheets justifying the change or deviation must be submitted with the manuscript copy. In submitting this information it is permissible to utilize the EJR format of figure 3. Request for deviation shall be prepared together with a separate engineering analysis in accordance with DOD-STD-480.

4.8 TRS identification and issue indicator. Each TRS shall be assigned a publication identification number at the time of acquisition in accordance with the procedures of the acquisition activity. For Navy activities, the publication numbers shall be assigned as specified in the Technical Manual Identification Numbering System (TMINS). If the TMINS has not been assigned, completion and submittal of NAVSEA Form 4160/5 by the preparing activity shall be considered an essential element of the manuscript copy submitted. An appropriate suffix for the TMINS shall be obtained from the TRS maintenance activity assigned principal interest in the document. The word "original" shall be located at the lower inboard edge of all pages of the basic TRS and superseding revisions. A change letter shall identify individual changed pages for control and status accounting purposes. The letter shall replace the word "original" as specified in MIL-M-38784 for changes to loose-leaf publications.

4.9 Applicability of TRS. Unless otherwise directed by the acquisition activity, a TRS shall be developed for specific system, equipment, or component configurations. The cover and title page shall reflect the identification.

4.10 Inspection and acceptance criteria. Items which are subject to wear, corrosion, erosion, and aging shall be examined and tested if such wear or other deterioration would result in failure to meet performance or reliability requirements. Specific acceptance criteria shall be furnished in the TRS for each such item. Those items requiring mandatory replacement shall be tabulated in the mandatory overhaul replacement parts list (see 5.12.1 and figure 4). The criteria shall be based on actual wear, corrosion, erosion, and aging data when such data are available. These criteria shall be based on assuring satisfactory performance for the planned period of service between refurbishments for the particular item. The criteria in many instances shall be based on engineering judgement, when wear or deterioration rates have not been established for shipboard operating conditions. Items other than shop stores (see 5.5.6(c)), mandatory replacement parts (see 5.8.3), or parts which rarely require replacement or whose cost approaches the value of the unit (see 5.12.2) shall be tabulated in the contingency material list (see 5.12.2 and figure 5), so that they are available if required.

4.10.1 Part identification and reference documents. Where specific parts are mentioned, the piece number and applicable reference document shall be written in the text of the TRS to minimize any possibility of misunderstanding.

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4.10.2 Technical acceptance and rejection criteria. When the technical manual includes charts of design clearances and tolerances, this information may be used as a starting point in specifying acceptance criteria. The following list illustrates typical examples of items and applicable criteria of acceptability:

(a) Bushings or bearings and mating shafts. Clearances or dimensions, surface conditions, and material requirements.
(b) Pump wear rings or rotors. Clearances or dimensions.
(c) Thrust bearings. End play.
(d) Gear trains. Backlash, contact pattern, and alignment.
(e) Pressure-containing parts. Wall thickness (thinnest area, erosion path, or both) and leakage rate.
(f) Couplings and shafts. Alignment and clearance.
(g) Piping, valves, fittings. Thickness, leakage rate, and proper material.
(h) Operating mechanisms. Backlash or play.
(i) Cylinders and pistons. Clearances, diameters, and leakage rate.
(j) Valves, disks, and seats. Dimensions for important parts, seat angles, seat rings, ball diameters, and proper material for replacement parts.
(k) Electrical wire. Insulation resistance.

4.11 Planned overhaul/repair record. The planned overhaul/repair records shall be as follows:

(a) Provide a permanent record of the overhaul/repair tests and inspections.
(b) Define items and areas requiring work.
(c) Provide data for future overhaul planning.
(d) Serve as a basis for determining the adequacy of the TRS for use on subsequent overhauls or refurbishments.
(e) Serve as a library of information which may be used in analyzing post-overhaul malfunctions.

The TRS shall require recording vital data, overhaul conditions, and TRS discrepancies. Instructions for the forwarding of planned overhaul/repair records shall be specified in chapter 1 (general information section of the TRS), and self-addressed forms shall be provided as directed by the acquisition activity.

4.11.1 Examination, test and repair action record (ETRAR). The planned overhaul/repair record requirement to record vital data and TRS discrepancies shall be accomplished through the preparation and completion of an ETRAR. The ETRAR shall summarize dimension, tolerance, and clearance parameters to complete inspections, repairs, and tests as specified in the text of the TRS. Each instruction to record data in the text of the TRS shall have an entry in the ETRAR. The requirement to record readings shall be identified in the text and ETRAR. A sample of an ETRAR is shown on figure 6.

4.11.2 Material condition/overhaul report. The planned overhaul/repair record requirements to document the overhaul condition shall be accomplished through the preparation and completion of the material condition/overhaul report (see figure 7). This record shall provide data on the general condition of the

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unit undergoing overhaul. The material condition/overhaul report also shall provide information on the condition of mandatory replacement parts, whether all acceptance criteria were met, and whether there was any approved deviation. Directions for completing the form shall be included in chapter 1, general information section of the TRS.

4.11.3 Technical manual deficiency evaluation report (TMDER). The requirement to provide feedback on TRS discrepancies shall be accomplished through the preparation and completion of the TRS TMDER (see figure 8). Directions for completion and use of the form shall be included in chapter 1 of the TRS.

4.12 Quality program. The preparing activity of the TRS shall prepare and maintain a technical manual quality assurance program and may be required to develop a validation plan, validation certificate, and verification incorporation certificate in accordance with MIL-M-85337 acceptable to the acquisition activity as specified in the contract or order (see 6.1 and appendix).

4.12.1 Quality program parameters. The quality program shall be delineated in accordance with the quality assurance provisions of MIL-M-15071 to assure that adequate and accurate TRSs are prepared and furnished in accordance with the requirements of this standard.

4.12.2 Government acceptance. Acceptance of the quality program by the acquisition activity is not intended to preclude additions, refinements, or changes to the written quality program where evidence indicates the need for such action by the acquisition activity, nor does it compromise the Government's right to perform quality assurance audits.

4.13 Level of writing, text and illustration development. The level of writing and development of text for the TRS shall be in accordance with the requirements of this standard, and the language style and writing requirements shall be in accordance with DOD-STD-1685 and MIL-M-38784, as applicable. The text of the TRS shall be specific, direct, concise, and clearly worded to be readily understandable by repair personnel and inspectors who have had appropriate training and working experience with the level of repair and overhaul of similar equipment. The U.S. Government Printing Office Style Manual shall be used as a guide for word capitalization, punctuation, compounding of words, use of numerals in text, and the spelling of nontechnical words.

4.13.1 Abbreviations and acronyms. Use of abbreviations and acronyms shall be held to a minimum, and each shall be defined the first time the abbreviation or acronym appears in each chapter. Thereafter, within each chapter, the abbreviation or acronym can be used. Abbreviations used shall be in accordance with MIL-STD-12.

4.13.2 Use of "shall", "will", "should", and "may". Use "shall" whenever a manual expresses a provision that is binding. Use "should" and "may" whenever it is necessary to express non-mandatory provisions. "Will" may be used to express a declaration of purpose. It may be necessary to use "will" in cases where simple futurity is required (for example, "Power for the meter will be supplied by the ship.").

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4.13.3 Accessibility and cross referencing. Cross referencing of technical information in text and illustrations shall provide rapid access to all data. When possible, text data tables and illustrations shall be located on the same or adjacent sheets. The user shall be able to proceed through the required action without undue interruption. Referencing shall be in accordance with the requirements of this standard.

4.13.4 Reference documents. Referencing shall be kept to a minimum and should be limited to Government-approved documents such as military specifications and standards, approved industrial standards, technical manuals, drawings, engineering change data, and other approved material which will enhance the clarity and support the repair requirements and processes delineated in the TRS. Where commercial documents are the only suitable reference material available, approval will be specified in writing by NAVSEA Life Cycle Manager (LCM) upon request of the TRS preparing activity. Strong justification shall be provided. Referencing shall conform to the following:

(a) Where a small amount of information is needed (less than two pages), the applicable material shall be extracted verbatim.
(b) Where a large amount of material is required (greater than two pages), the applicable material shall be referenced fully, including the publication number and the title. An example is NAVSEA 0900-LP-038-6010 Chrome Plating. Reference to alternative repair procedures and to drawings required to refurbish parts is acceptable, if that refurbishment information is not normally required.
(c) Where reference is to an entire content of another document, the reference shall be only to the title of the publication and the publication identifying number. It is not considered necessary to repeat quality control standards or other approved routine procedures which have been addressed during specialty training. For example, the requirements of MIL-STD-278 (Welding) or NAVSEA 0948-LP-045-7010 (level I material) should not be repeated but shall be invoked through reference.

4.13.4.1 Text reference placement. Reference within the text shall conform to the following:

(a) When a reference applies to one item within a sentence, place the reference parenthetically immediately after the item being referenced, for example: "The test set (see figure 1-1) comprises a main panel (see figure 1-2) and a power supply (see figure 1-3)." When a reference applies to an entire sentence, place the reference at the end of the sentence: "This characteristic causes the first portion of control movement to be less effective (see figure 3-3)," or show the reference in a complete sentence: "The procedure for making the adjustment is shown on figure 3-23."
(b) When a reference applies to an entire paragraph or paragraphs, place the reference after the paragraph heading, for example, "3-2 Condenser Tube Nest Arrangement (see figure 3-56)."

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4.13.4.2 Enclosures. Enclosures may be used to include closely related material which is not an inherent part of the TRS. The enclosures, identified in full, shall be listed by numbers (for example, (1), (2), and (3)) in the order that they are mentioned in the text. Enclosures shall be inserted immediately after the last appendix. The page designation shall be "Enclosure _____, page _____ of _____".

4.13.5 Alternate source data approval. When the technical data required to develop a TRS are incomplete or unavailable, the necessary data shall be generated by the process of reverse engineering (see 3.13). Prior to approval of a TRS developed by this process, a unit overhauled in accordance with it shall be tested and evaluated in accordance with the applicable equipment military specification requirements to assure performance characteristics are satisfied.

4.13.6 Copyrights. Commercial copyright and proprietary material shall not be incorporated unless essential to satisfy the TRS objectives. When it is necessary to include this material, it shall be clearly identified and the following warning statement included on the TRS cover sheet and title page:

"This document contains copyright and proprietary materials. Infringement of copyright and proprietary material may violate existing Federal laws and statutes and result in criminal penalties, imprisonment, or removal from office."

The preparing activity shall be responsible for resolving questions concerning commercial copyright and proprietary material and obtaining written permission of the data owner and acquisition activity for utilization of the data and restrictions on its use.

4.13.7 Illustrations, drawings, and sketches. Illustrations shall be provided to convey essential information and to support instructions in the TRS. Illustrations shall be used where they will improve clarity and reduce the volume of written descriptive matter (see figure 9). Illustrations shall conform to the following:

(a) Illustration placement. Each illustration shall follow, as closely as possible, its first reference in the narrative text. However, when an illustration is extensive and requires a foldout sheet, such as for a large diagram, it shall be located at the end of its relative discussion in the narrative text or at the end of a chapter. Foldup-foldout (map fold) illustrations shall be prohibited. Preparation shall be follows:

(1) Maximum size foldouts shall be 279 mm by 114 cm (11 by 45 inches) with 254 by 902 mm (10 by 35-1/2 inch) printable area including running heads, figure numbers, titles, and pagination.
(2) Foldouts shall be prepared as right-hand pages printed on one side only.
(3) Aprons (216 mm or 8-1/2 inches) shall be provided on the binding edge of all foldouts.
(4) Color or halftone artwork shall not be used without prior approval of NAVSEA.

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(b) Illustration titles. When necessary, for reference purposes, illustrations shall be assigned figure numbers and titles. The title shall identify the contents or purpose of the illustration in such a manner as to distinguish that illustration from others in the TRS. The title shall follow the number (separated by a period and two spaces) on the same line. The complete caption shall be centered below the illustration. If the figure title exceeds one line, the second line shall be flush left under the first word of the title. Single spacing shall be used. The word "Figure" and each principal word in the title shall be initial capitals. The complete title for sectionalized foldout illustrations shall be placed flush right on the last segment.

(c) Material or parts list. A material or parts list shall be included with each illustration. Only those parts referenced in the text shall be identified on the drawing, or when specifying "disassembly in accordance with disassembly sequence numbers," all numbered parts shall be listed. Text and illustrations shall complement each other to communicate the required information. Nomenclature shall be consistent throughout the TRS.

(d) Parts identification. Parts shall be identified with the technical manual part number. Where it is sufficient to specify "Disassemble by removing parts in accordance with the disassembly sequence number," sequence numbers shall appear in parentheses after the technical manual number on the figure. Only those parts identified with part numbers on appropriate figures shall be listed in the parts list. Parts shall be listed in order of sequence numbers when applicable. The parts list shall consist of the following (with headings from left to right):

(1) Technical manual number.
(2) Disassembly/assembly sequence number.
(3) Item name.
(4) Quantity in parentheses.

(e) Illustrations shall identify key parts locations with callouts attractively placed around the drawing as shown on figure 9. Care shall be taken not to introduce clutter and distractions.

4.13.7.1 Examples of illustration types. The following are examples of types of illustrations and shall be provided, commensurate with the complexity of the equipment covered by the TRS:

(a) Isometric exploded views.
(b) Sectional views.
(c) Test set-up diagrams.
(d) Assembly, disassembly, reassembly, and fabrication drawings.
(e) Schematic diagrams, block diagrams, and timing circuit diagrams.
(f) Wiring diagrams.
(g) Item location illustrations.

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(h) Engineering drawings.
(i) Gearing and linkage diagrams.
(j) Illustrated parts breakdown.

4.13.7.2 Illustration selection criteria. Exploded view isometric drawings shall be used whenever possible. Sectional views or item location drawings should be used to indicate where critical measurements shall be taken or when the information presented is easier to understand when depicted as a cross-sectional view. Parts should be numbered with the equipment's technical manual part numbers. A disassembly/reassembly sequence number, in parentheses, shall follow the technical manual part number, as applicable. Both numbers shall also be included in an accompanying parts list including part names and quantities. Sectional drawings should not be used when isometric drawings are provided.

4.13.7.3 Acceptability of drawings and illustrations. Engineering drawings and illustrations which were not prepared primarily for TRS illustration purposes are acceptable if the copy print is legible, reproducible, and readable when reduced to TRS size. Irrelevant material that would not be required in accomplishing the TRS shall be removed.

4.13.8 Warnings, cautions, and notes. Warnings, cautions, and notes shall be short, concise statements used to emphasize critical or hazardous procedures or situations.

4.13.8.1 Warnings and cautions. WARNING and CAUTION statements shall be included for emphasis in overhaul, repair, and testing procedures in accordance with MIL-M-38784 and the following:

(a) A WARNING statement shall be used to call particular attention to a step of a procedure which, if not strictly followed, could result in serious injury or death of personnel.
(b) A CAUTION statement shall be used to call attention to a step of a procedure which if not followed, could result in damage to, or destruction of, equipment.
(c) WARNING and CAUTION statements shall immediately precede the text of those procedural steps where failure to strictly follow the particular step of a procedure would be likely to result in personnel injury or death, or damage to or destruction of equipment. The WARNING and CAUTION shall state the consequences of failure to strictly follow the procedural step to which it applies.
(d) The use of WARNING or CAUTION statements shall be limited to significant cases. Extensive use of these statements will make overhaul personnel apathetic of them and thus reduce their value.
(e) WARNING or CAUTION statements shall not contain procedural steps or alternate procedural steps, nor shall they be numbered.
(f) When a WARNING or CAUTION consists of two or more paragraphs, the heading "WARNING" or "CAUTION" shall not be repeated above each paragraph. The format of either text (for example, indentation or italics) should suffice to identify the following paragraphs.
(g) If it is necessary to precede a paragraph by both a WARNING and CAUTION, the WARNING shall precede the CAUTION.

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4.13.8.2 Health hazards. Procedures prescribed for the overhaul or testing of equipment shall be in accordance with the safety standards established by latest revision to OPNAVINST 5100.23. Warnings and cautions shall be used when hazardous chemicals, adverse health factors in the environment, or use of dangerous equipment cannot be avoided. A list of personnel protective devices and clothing shall be specified.

4.13.8.3 Notes. Notes present an essential precondition statement which shall be highlighted and shall be as specified in the following:

(a) Notes shall not be used with or in lieu of procedural steps.
(b) Notes shall be confined for use in narrative or illustrative nonprocedural data. Notes shall never be used in lieu of WARNINGS or CAUTIONS, nor shall they be used to modify an action.
(c) Notes shall precede the procedural step to which they refer.

4.13.9 Tables. Tables shall be in accordance with the following:

(a) Center the word "Table," the applicable number, and the title above the head rule.
(b) Initial capitalize "Table" and each principal word in the title.
(c) Capitalize each principal word in column heads.
(d) Single space entries within the table.
(e) Related entries in different columns shall be aligned.
(f) Entries within columns shall be aligned as follows:

(1) For decimal data, decimal points shall be aligned.
(2) For mathematical notation, multiplication signs shall be aligned.
(3) All other numeric data shall be aligned flush right.
(4) Alphabetic or alphanumeric data shall be aligned flush left.

(g) Carryover lines shall be indented two spaces.
(h) The units of measurement shall be specified in row or columnn headings.
(i) The row entries in tables shall be arranged in groups of up to five rows. Groups shall be separated with white space if no entries are blank and with light horizontal lines if any entries are blank. At least 25 percent of the area within tables and between columns and groups of rows shall be white space (see DOD-STD-1685).

4.13.9.1 Ruled lines on tables. A horizontal rule shall be placed at the beginning (head) and at the end (foot) of a table. A horizontal rule shall also separate column heads from data entries. Vertical rules may be drawn between columns where required for clarity. The rule shall not be placed to the left of the first column or to the right of the last column to box in the table. The closing rule shall be placed at the foot of a table on the last page of the table.

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4.13.9.2 Continued material for tables. When a table requires more than one page, the number and title shall be repeated at the head of columns on all following pages of the table, followed by "(Contd)." Column heads shall also be repeated. When information opposite an item is continued, the item shall be repeated, followed by "(Contd)."

4.13.9.3 Footnotes on tables. Numbering of footnotes applicable to tables shall be independent of footnotes to the text (see 4.13.11.1). The reference shall be indicated by consecutive Arabic numerals in superscript position within each table. (Superscript lowercase letters, asterisks, or other designation may be used where numbers could cause confusion.) The footnotes, which shall be kept to a minimum for clarity, shall be placed immediately below the table in which they are referenced. When a table is continued onto other pages, all footnotes shall be placed at the bottom of the page on which they are referenced. Table notes and footnotes shall be indented five spaces and carryover lines shall return to the left margin.

4.13.10 Marginal copy. Marginal copy (generally the TMIN, page number, and change designator) shall be placed outside the 178 by 229 mm (7 by 9 inch) image area reserved for narrative text, tabular data, or illustrations, but within the 178 by 254 mm (7 by 10 inch) printing area of the page on standard 216 by 279 mm (8.5 by 11 inch) documents. When applicable, marginal copy may also consist of the security classification, page content, equipment identification, figure number, and figure title. Line illustrations, full page or foldout, that have been prepared in exact printing size shall have the marginal copy mounted thereon.

4.13.10.1 Page number location. The page number shall appear in the lower-outer corner of each page. When the information on a page is placed head-to-foot (such as in an appendix), the page number shall be placed in the lower right-hand corner relative to the orientation of the text on that page.

4.13.10.2 Blank page numbering. Blank pages shall be assigned numbers, but such numbers shall be located on the preceding pages, (for example, if page 10 of Chapter 1 is blank, page 9 shall show both numbers: 1-9/(1-10 blank).

4.13.11 Numbering of pages, paragraphs, tables, and illustrations. Numbering shall be consecutive within each chapter, with even numbers used for left-hand pages. Two-part Arabic numerals shall be used, separated by a hyphen. The first number shall signify the chapter, and the second the order within the chapter of the page, paragraph, table, or illustration. For example, the numbers 2-17 signify Chapter 2; and page, paragraph, table, or illustration number 17 within that chapter. A chapter may contain a paragraph, table, and illustration identified by the same number. Decimal points may be used if it is appropriate to break paragraphs down, such as 2-17.1, and so forth. If a full-page illustration is appropriate at the beginning of a chapter, it shall be placed on a left-hand page and numbered "0", such as 1-0, 2-0, and so forth. Pages in each chapter shall be sequentially numbered, starting with "1" within each chapter. Page designations for the table of contents, list of illustrations, list of tables, and the safety summary shall be numbered sequentially starting with lowercase Roman numeral "i".

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4.13.11.1 Text footnote numbering. Footnotes, if required, shall be numbered consecutively within each chapter by superscript Arabic numerals. Footnotes to the text shall be placed at the bottom of the page on which they appear. Footnotes to tables shall be placed immediately at the bottom of the table in which they appear (see 4.13.9.3).

4.13.11.2 Headings. Numbers and titles for chapters, sections, appendices, alphabetical index, and so forth, shall be centered at the top of the first page of text for each. Chapters, sections, appendices, and the alphabetical index shall begin on a right-hand page. Section headings shall be centered immediately below the chapter heading.

4.14 Legibility. The copy shall be clear and readable. In addition, the copy shall be satisfactory for use in making 35 mm microfilm of each page in accordance with MIL-M-9868 or microfiche in accordance with MIL-M-38748. Letters, lines, and symbols shall be of a uniform contrast throughout the TRS. Layout shall conserve space without reducing the usability or clarity of material. Blank pages and spaces shall be avoided whenever possible. Spacing shall be used for best legibility and conservation of space. Double spacing of text within a paragraph in the final TRS, or similar wastefulness, shall not be allowed. Slight variations in the final TRS are permitted to avoid the following layout practices:

(a) The first line of a paragraph or a paragraph heading on the last line of a page.
(b) The last line of a paragraph at the top of a new page.
(c) Warnings, cautions, and notes divided so that the first lines (or headings) appear on one page and remaining lines on another.

4.15 Uses of appendices. Appendices shall be used to separate bulky information from the body of the TRS, when such separation will increase the clarity of the overall document. The appendix shall be referenced in the basic text of the standard, indicating the extent to which it applies. A statement shall appear at the beginning of the appendix to indicate its scope and limitations, and to assure its proper application and use. The following are examples of material which shall be placed in appendices:

(a) Mandatory overhaul replacement parts list (see figure 4).
(b) Contingency material list (see figure 5).

4.16 Covers and title pages. Covers and title pages for standard unclassified TRSs shall be prepared as shown on figure 10 and as follows:

(a) Type style shall be Univers, Medium Condensed, or equivalent. Type sizes indicate the height of capitalized letters. (One point is equal to 0.035 cm or 0.014 inch (see DOD-STD-1685).)
(b) A single master shall be prepared for production of both the cover and title page. Flaps can be used to cover the "Approved By" signature and publication date position. Cover the signature and date when printing front cover; lift flaps to print title page.

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(c) Spacing and placement of necessary information may be varied slightly according to the number and length of information items to be presented. The end product shall result in an attractive, well balanced cover and title page.
(d) The back cover shall be made of the same material, color, and quality as the front cover. Copy for the front and back cover of a classified TRS shall show the security classification, at the top and bottom.

4.16.1 Publication date. The publication date is the copy freeze date and shall be the day of the month in which the latest material included in the TRS was prepared. The day, month, and year shall be shown in that order on the title page only, below and to the right of authority notice. The publication date shall not be shown on the cover. The publication date of volumes that are composed of two or more parts shall be the same.

4.16.2 Paper stock. The TRSs shall be prepared on any good quality white paper stock 216 by 279 mm (8.5 by 11 inches) which is suitable for reproduction. Foldout pages conforming to the requirements of this standard and paper stock for printed manuals in accordance with MIL-M-15071 shall be used.

4.16.3 TRS binding. TRSs shall be loose-leaf bound with screw fasteners in pressboard, or equivalent binders, or placed in a 3-ring notebook.

4.17 Preparation for delivery. Packaging and packing of deliverable items acquired by this standard shall be in accordance with MIL-M-38784. Marking shall comply with security requirements. Deliverables shall be packaged to prevent damage in transit. Bulk shipment of TRSs for stock shall not be individually wrapped. Containers for deliverable data items shall be in accordance with MIL-M-15071, the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation for the specified deliverable product. On bulk shipment, interior packages and exterior shipping containers shall be marked with the following information for each item enclosed, except for shipment of an individual copy:

(a) Box _____ of _____ (enter numbers).
(b) TMIN.
(c) TRS title.
(d) Quantity.
(e) Contract or order number.

4.18 Permanent changes. A technical repair standard change notice (TRSCN) (see appendix) shall be used to propose, record, and transmit changes to a TRS, as required (see figure 11). The TRSCN shall be used as a cover sheet and letter of transmittal. The changes associated with the TRSCN shall be attached and shall constitute an integral part of the TRSCN. Urgent or other changes shall be handled in accordance with the standard procedures of the acquisition activity and the hardware systems command.

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4.18.1 TRSCN. The TRSCN shall contain the exact change in TRS paragraphs, figures, or other content that will be distributed to users if the TRSCN is approved. Modifications shall be prepared in final change page format together with a new cover sheet and title page. Page sizes and punching or drilling for TRS changes or revisions shall be the same as for the original basic TRS. Complete instructions for substituting the new pages shall be printed on separate cover sheets.

4.18.2 Incorporation of changed pages. Revised and re-issued pages shall be complete reprints of the front and back of pages, suitable for incorporation by removal of old pages and insertion of new pages. Portions affected by the change shall be indicated by a vertical symbol in the left-hand margin, adjacent to and encompassing all changed portions.

4.18.3 Changed page status. Each changed page of the TRS shall be identified with the TMINS assigned to the basic TRS. The change status of individual pages shall be indicated by the change identifier in the marginal copy ("running foot") at the inboard bottom of each changed or added page.

4.18.4 Changed page numbering. The changed pages furnished with a TRSCN shall be numbered the same as the superseded pages. If it is necessary to replace one page with more than one, the additional pages shall carry the same number as the affected page, plus a lower case suffix letter in alphabetical order. For example, to change page 5, the changed page numbers would be 5, 5a, 5b, and so forth. If a page is deleted, that number shall be omitted in the current page sequence and shall be so reflected in the list of effective pages (LOEP) (see 5.4.2).

4.19 Revisions. A revision of a TRS is a re-issue of a complete TRS. The TRS revision shall be prepared, issued, and identified as directed herein (see appendix). Each revision shall incorporate all outstanding approved changes against the previous issue, as well as approved changes proposed by the TRSCN that creates the need for revision. A complete revision supersedes a basic TRS when a major portion of the basic TRS is designated inadequate because of obsolescence, method of presentation, arrangement, or any other reason detrimental to the user (see appendix). The acquisition activity will determine and approve the extent of revision required.

4.20 Approval. TRSs shall be approved by the NAVSEA LCM. Once a TRS has been approved by the acquisition activity, it shall not be modified without further approval.

4.21 Precedence. In the event of conflict between this standard and the authorized contract, Statement of Work, Military Interdepartmental Procurement Requests (MIPR), or other documents referenced in the Contract Data Requirements List (CDRL), the contract and attendant technical manual contract requirement or SEATASK requirement (TMCR/TMSR) shall take precedence.

4.22 Rejection criteria. Failure of the preparing activity to conform to the requirements herein to the degree cited below shall require the preparing activity to modify the subject TRS to correct the deficiencies.

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(a) Technical accuracy and inconsistencies. Failure of all information to exactly represent the equipment or system being described and to be consistent with descriptive terms.
(b) Omission of data. Omission of any applicable data elements or inspection and verification (I&V) points.
(c) Evidence of validation. Failure to provide adequate evidence of validation.
(d) Workmanship. Significant typographical and grammatical errors.
(e) Change requirements. Failure to include changes or corrections required as conditions of approval of the manuscript.

5. DETAILED REQUIREMENTS

5.1 TRS organization and divisions. Technical subject matter in a TRS (see appendix) shall be presented as described in 5.1.1 through 5.1.5. For data requirements, see appendix.

5.1.1 Volumes. Each TRS shall be formatted into two separate parts, according to required subject matter as specified in 5.2 and shall be identified by the NAVSEA TMIN title and subtitles indicative of volume content. TRS prime titles for HM&E equipment shall be consistent with the official Expanded Ship Work Breakdown Structure (ESWBS) nomenclature. Separately-bound volumes shall not exceed 76 mm (3 inches) in thickness.

5.1.2 Parts. The TRS shall be divided into two parts. Part 1 shall contain all overhaul procedures, requirements, and drawings necessary to perform the overhaul. Part 2 shall provide for the recording of all readings specified in part 1.

5.1.3 Chapters. The TRS shall be divided into chapters as specified in 5.2 to provide logical content arrangement. Chapters shall start on a right-hand page and shall be identified by an Arabic numeral on the heading page. Pages within a chapter shall be identified by a chapter number, a hyphen, and successive page numbers.

5.1.4 Sections. When multiple-coverage in chapters of TRSs for compound items are used, a chapter sectioning technique may be employed. In this instance, a section that reflects a separate breakout item or separate APL-identified unit shall start on a new right-hand page. Roman numerals (capitalized) shall be used to number sections consecutively within the chapter to which they pertain.

5.1.5 Paragraphs. Text shall be divided into primary paragraphs and subordinate paragraphs. Subordinate paragraphs may be identified as first subordinate, second subordinate, and third subordinate. Paragraphs may also be divided into procedural steps. Procedural steps may be divided if necessary. Decimal numbering specified in 5.1.5.6 shall be used.

5.1.5.1 Paragraph headings. Informative paragraph headings shall be used to identify subject matter. Paragraph headings shall be identified as primary sideheads, first subordinate sideheads (secondary), and second subordinate sideheads (tertiary). Each paragraph heading (sidehead) shall be descriptive of the content of the material which it heads. Each paragraph sidehead shall start with a significant word. Sideheads shall be as follows:

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(a) Headings shall be confined to seven words for normal usage and not more than ten words.
(b) Headings shall be unique within any major portion of the TRS.
(c) Periods shall follow paragraph headings.
(d) Numbered paragraphs shall start on the left margin. Paragraphs without sideheads shall be handled as procedural steps.

5.1.5.2 Primary sideheads. Primary paragraphs shall normally be used to divide text within chapters or sections into two or more main portions. There shall be at least two primary paragraphs in each chapter or section. A primary paragraph begins with a primary sidehead and shall be prepared as follows:

(a) All capitals.
(b) Stand alone (no text on the line).
(c) Underscored or prepared in bold face type.

5.1.5.3 First subordinate sideheads. First subordinate paragraphs shall be used to divide text under primary headings into two or more subjects. There shall be at least two first subordinate paragraphs when a primary subject is subdivided. First subordinate paragraphs begin with first subordinate sideheads which shall be prepared as follows:

(a) All capitals.
(b) Underscored.
(c) With text being run-in. (Text and title shall be separated by a period and two spaces.)

5.1.5.4 Second subordinate sideheads. Second subordinate paragraphs shall be used to further subdivide material under first subordinate paragraphs. Second subordinate paragraphs normally begin with second subordinate sideheads which shall be prepared as follows:

(a) Upper and lowercase. (First letter in each word capitalized.)
(b) Underscored.
(c) With text being run-in. (Text and title shall be separated by a period and two spaces.)

5.1.5.5 Third subordinate sideheads. Third subordinate paragraphs shall be used to further subdivide material under second subordinate paragraphs. These numbered subordinate paragraphs begin with subordinate sideheads which shall be prepared the same as second subordinate sideheads. Breakdown beyond the third subordinate shall not be used.

5.1.5.6 Procedural steps. Procedural steps or checklists shall be used to provide step-by-step instructions for disassembly, reassembly, or overhaul of a system or equipment. The steps may be divided into substeps. Procedural steps or checklist items shall be indented and presented in a block format as shown below. (Typography and numbering of primary and subordinate sideheads shall complement heading-order requirements.)

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Order of Heading	Margin
(1)	3-1 PRIMARY SIDEHEAD.
(2)	3-1.1 FIRST SUBORDINATE SIDEHEAD.
(3)	3-1.1.1 Second Subordinate Sidehead.
(3)	3-1.1.2 Second Subordinate Sidehead. XXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.
	a. First procedural step. (XXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.)
	(1) First subordinate level procedural step. (XXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.)
	(a) Second subordinate level procedural step. (XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.)
	3-2 NEXT PRIMARY SIDEHEAD.
	3-2.1 FIRST SUBORDINATE SIDEHEAD. XXXXXXXXXXXXXXXXXXXXXXXXX.
	a. First procedural step. (XXXXXXXXXXXXXXXXXXXXXXXXX.)
	b. Second procedural step. (XXXXXXXXXXXXXXXXXXXXXXXXX.)

5.1.6 Appendix. Appendices shall be identified by capital letters ("APPENDIX A" and so forth). They shall immediately follow the last chapter of the TRS. Appendix headings shall be centered. Pages, paragraphs, illustrations, and tables for appendices shall be identified by Arabic numerals, preceded by the capital letter of the appendix (for example: A-17 shall be the 17th page in appendix A).

5.2 Subject matter presentation. For an effective and efficient method of performing the class B overhaul of the items covered in the document, part 1 of the TRS shall provide the minimum instructions for critical procedures, fewest possible tests and maintenance examinations, supporting illustrations, lists of items to be replaced, special tools, critical operations and checks during reassembly, minimum quality control checkpoint requirements, and specific post-overhaul acceptance tests or inspections. Part 2 of the TRS shall provide for the recording of all data requirements contained in part 1. Part 2 shall be structured to complement part 1. The TRS shall be formatted in accordance with 5.2.1 through 5.2.3.

5.2.1 Technical subject matter. The technical subject matter shall follow the front matter and shall be arranged in chapters. Each chapter shall consist of one or more sections which are divided into main paragraphs and subparagraphs. Sections and major (numbered) paragraphs shall each have a brief title describing the content or action required (see figure 12). The level of detail shall be geared to the journeyman level mechanic.

5.2.2 Components. Subject matter with major or minor components shall be divided such that each item will be addressed in a separate chapter to promote the work-unit, work-package concept.

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5.2.3 Subassemblies. Minor subassemblies shall be divided into separate sections and included with their major subassemblies to promote the workgroup breakdown structure of authorized overhaul facilities.

5.3 Format. The general divisions and presentation of data for a compound item shall be a two-part volume as follows (see figure 12).

5.3.1 Part one. Part one of the volume shall contain the following:

(a) Front cover and title page (see 4.16 and figure 10).
(b) Front matter (see 5.4).
(c) Chapter 1 - General Information (see 5.5).
(d) Chapter 2 - Shipboard Removal (see 5.6).
(e) Chapter 3 - Compound Item Disassembly (see 5.7).
(f) Chapter 4 - Major Component (for example, pump); Disassembly, Inspection, Repair, Reassembly, Test (see 5.8).
(g) Chapter 5 - Major Component (for example, driver); Disassembly, Inspection, Repair, Reassembly, Test (see 5.8).
(h) Chapter 6 - Unit Assembly (as applicable) (see 5.9).
(i) Chapter 7 - Shop Unit Test (as applicable) (see 5.10).
(j) Chapter 8 - Shipboard Installation, Alignment, and Testing (see 5.11).
(k) Appendix A - Mandatory Overhaul Replacement Parts List (see 5.12.1).
(l) Appendix B - Contingency Material List (see 5.12.2).
(m) TMDERs (see 5.13).
(n) Back cover (see 4.16).

5.3.2 Part two. Part two of the volume shall contain the following:

(a) Front cover and title page (see 4.16 and figure 10).
(b) Front matter that describes content (see 5.4) and instructions for collecting and disposing of the data package (see 5.5.3).
(c) Material condition/overhaul report (see 5.12.3).
(d) Appendix C (see 5.12.4).
(e) Additional appendices (see 5.12.5).
(f) Enclosures (see 5.12.6).
(g) TMDERs (see 5.13).
(h) Back cover (see 4.16).

5.4 Front matter. Front matter shall consist of the following in the order specified:

(a) Cover and title page.
(b) List of effective pages.
(c) Change record.
(d) Table of contents.
(e) List of illustrations.
(f) List of tables.
(g) Safety summary.

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5.4.1 Cover and title page. The cover and title page shall be as specified in 4.16 and figure 10.

5.4.2 List of effective pages (LOEP). A LOEP shall be prepared in accordance with figure 13. This page shall back up the title page and shall be identified by the letter "A" in the lower left-hand corner. When additional space is required, "B" and "C" pages shall be added as right- and left-hand pages, respectively. The LOEP shall be a complete list of all TRS pages, including title page, "A" page, blank pages, deleted pages, added pages, and foldout pages. The LOEP shall be updated to incorporate each TRS change. The listing shall be held to a minimum by grouping numbers, where applicable. The page numbers for a blank page and the printed side of the sheet shall be listed as separate numbers, even though a double number shall be shown in the "Change No." column. The words "Deleted" or "Blank" shall be placed alongside the page number of pages so affected. For multivolume TRSs, a LOEP covering all volumes shall be prepared and included in volume 1. Each of the TRS volumes shall also include a listing of pages provided in that particular volume.

5.4.3 Change record. A change record shall be prepared in accordance with figure 14. The change record page shall be a right-hand page and shall be inserted immediately following the list of effective pages. The page designation shall be "Change Record/(Change Record-2 blank)" to indicate the back side of the page is blank.

5.4.4 Table of contents. A table of contents listing all chapters, sections, and numbered major paragraphs in the same order and with the exact titles used in the text shall be placed at the beginning of each TRS immediately behind the change record. Layout shall be as shown on figure 15. Lower case Roman numerals shall be used to identify the page numbers of the table of contents. When multivolume TRSs are authorized, volume 1 shall contain a table of contents for all volumes in the set. Each volume shall contain its own table of contents.

5.4.5 List of illustrations. Illustrations identified in the text (including charts and graphs assigned figure numbers) shall be included on a list of illustrations as part of the table of contents. A typical layout is shown on figure 16.

5.4.6 List of tables. Tables identified in the text shall be included in a list of tables as part of the table of contents. A typical layout is shown on figure 17.

5.4.7 Safety summary. TRSs containing personnel safety warnings shall include a safety summary in accordance with figure 18 in each affected volume to highlight the potential hazards. It shall include all general warnings (high voltage warning, warning against servicing alone, safety clothing requirements, and resuscitation). The safety summary shall comprise the final section of the front matter immediately ahead of the technical content.

5.5 Chapter 1 - general information. This chapter shall provide a general overview of the data presented in the TRS. The chapter shall include the information specified in 5.5.1 through 5.5.8 and shall be identified with numbered paragraph headings as applicable (see 5.1.5.1).

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5.5.1 Scope. The scope shall include a brief summary of the purpose, contents, and arrangements of the TRS, including appendices. It shall include a statement that conformance to the TRS is required of designated commercial and Government activities authorized for overhaul of the items covered in the TRS in order to meet specified class B overhaul criteria.

5.5.2 Points of contact. This paragraph shall state that the TRS takes precedence over all other technical documents. It shall further state that the shop supervisor or design division should be contacted regarding problems associated with use of the TRS.

5.5.3 Overhaul conformance. This paragraph shall state that compliance with the applicable chapters of the TRS is mandatory and that, upon completion of overhaul, equipment shall conform to the specified performance requirements. This paragraph shall also contain statements:

(1) Specifying that configuration changes are not authorized without NAVSEA approval.
(2) Regarding visual inspection requirements of replacement parts and that the TRS does not specify replacement part testing.
(3) Specifying that parts listed in the mandatory overhaul replacement parts list (appendix A) are listed based upon failure history and cost to repair or replace, and these parts shall be replaced regardless of condition; therefore, this material shall be on hand prior to performing a class B overhaul in accordance with the TRS.
(4) Specifying that the contingency material list (appendix B) reflects those parts which may be necessary when conducting the overhaul, and acquisition of these contingency parts shall be based upon the experience of the overhauling activity.
(5) Directing the TRS user to take and record specified readings, test results, inspection and examination results, and corrective action; specifying that these data be forwarded to the cognizant TRS maintenance activity.
(6) Specifying that data be recorded at mandatory "I" and "V" points; designating the individual to sign off on all mandatory "I" and "V" points; and specifying that the local activity may add additional "I" and "V" points.

5.5.4 References. A tabular listing shall be developed to present all documents, publications, product engineering drawings, schematics, specifications, standards, engineering changes data, and other references in the specific technical requirements chapters (chapter 2 through appendices) of the TRS. Applicable documents shall be listed in the order they are referenced in the text.

5.5.5 Unit/assembly/subassembly. This paragraph shall identify the repairable items covered by the TRS. Items shall be identified by nomenclature, model number, APL, FSCM, and manufacturer's part number. These data shall be listed in tabular form. This paragraph shall also provide a tabular list of the ship or machinery alterations which have been incorporated into the TRS and the hull numbers to which they apply. If no ship or machinery alterations have been incorporated into the TRS, a statement to that effect shall be included.

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5.5.6 Parts replacement. Subparagraphs in this topic area shall be developed to specify the following:

(a) The local manufacture or fabrication of certain restricted parts is strictly forbidden. Due to the critical nature and stringent manufacturing requirements of these parts, NAVSEA has limited the manufacture of these restricted parts to qualified sources. A printout of restricted parts may be obtained from NAVSEA Logistics Support Activity, Mechanicsburg, PA (Code 51). In addition, those parts which are also mandatory replacement or contingency spare parts shall be highlighted by a double asterisk in appendices A or B, and in the text as applicable.

(b) Parts examined which do not meet acceptance criteria may be repaired (if not on appendix A or a restricted part) or shall be turned in to the supply system or disposed of utilizing current procedures and documentation. Local manufacture of nonrestricted parts can be accomplished when acquisition is delayed; however, these parts shall conform to drawing specifications and source inspections.

(c) Shop stores items such as gaskets and packing material, locknuts and lockwashers, cotter pins, keys, and so forth, shall not be listed in the mandatory overhaul replacement parts list or contingency material list but shall be replaced. This material shall be available in order to accomplish the planned overhaul maintenance.

5.5.7 Special facilities, test equipment, skills, and test conditions. This paragraph shall note the personnel skills and special equipment or facilities required to accomplish the equipment overhaul covered by the TRS. It shall include the following information as required:

(a) The special facilities required to perform the overhaul shall be identified and prescribed. The information shall be normally listed in a tabular form.

(b) Special test equipment or acceptable alternative items and repair aids required (specify quantities) in the examination, repair (including alignment, adjustment, and troubleshooting), and performance or acceptance testing of the items covered by the TRS shall be identified and described. A tabular listing is preferred.

(c) Knowledge, skills, and abilities required to accomplish the equipment overhaul shall be defined. This information shall be presented in a brief narrative form or a tabular listing.

(d) Special work station environmental conditions required in the overhaul process shall be specified, other than atmospheric pressure, specific temperatures, relative humidity, or clean rooms. This information may be detailed in table format.

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5.5.8 TRS users comments procedure. A brief instruction to the TRS user shall be developed specifying that comments or recommendations concerning the TRS are to be forwarded for consideration using the TMDER system. The following statements shall be included:

(a) Errors, omissions, discrepancies, and routine suggestions for improvement to NAVSEA TRSs shall be reported to the Commanding Officer, Naval Ship Weapon Systems Engineering Station (NAVSHIP-WPNSYSENGSTA) (Code 5700), Port Hueneme, CA 93043 on NAVSEA Technical Manual Deficiency/Evaluation Report, NAVSEA Form 9086/1(4-84). To facilitate such reporting, three copies of Form 9086/1 are included at the end of this TRS. Feedback comments shall be thoroughly investigated and originators will be advised of action resulting therefrom. Extra copies of NAVSEA Form 9086/1 may be requisitioned from the Naval Publications and Forms Center (NPFC), Philadelphia, PA 19120.

(b) Urgent priority TRS deficiencies shall be reported by Naval message with transmission to NAVSHIPWPNSYSENGSTA, Port Hueneme, CA; the designated In-Service Engineering Agent (ISEA) (specify by short title from the U.S. Navy Plain Language Address Directory); the TRS maintenance activity; and the commodity life cycle manager (specify the NAVSEA Code) as multiple addressees. Local message handling procedures shall be used. The TRS by title and TMIN shall identify each such message processed.

5.6 Chapter 2 - shipboard removal. This chapter shall provide information such as procedural, inspection, repair, data recording and test criteria with supporting appendix pages, warning and caution statements, and sketches in accordance with the applicable requirements specified herein for the following:

(a) Unit boundary identification, addressing overhaul/considerations for complete or partial unit overhaul/removal.
(b) Draining unit.
(c) Insulation removal (with appropriate asbestos hazard warning).
(d) Interference, interface, and piping removal (with appropriate asbestos hazard warning).
(e) Unit installation site (foundation, interface, and so forth):

(1) Preparation.
(2) Cleaning.
(3) Inspection.
(4) Repair.
(5) Blanking-off, tagging, and so forth (including applicable tag out requirements).
(6) Removal routes (where specific removal routes have been developed, they shall be referenced as required in 4.13.4).

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5.7 Chapter 3 - unit disassembly. This chapter of the TRS shall specify the requirements for disassembly of the major components of the unit in order to facilitate the work breakdown structure of a compound unit, as broken down into repair sections in chapters 4, 5, and so forth, as required. Procedural and data recording requirements with supporting appendix pages, warning and caution statements, and sketches in accordance with the applicable requirements shall be provided for the following:

(a) Major unit boundary identification.
(b) Coupling separation/interface and piping removal, electrical disconnection.
(c) Major unit removal.

5.8 Chapter 4 (and others as required) - item overhaul procedures. These chapters shall contain procedures, minimum test and inspection requirements, processes, methods, and data deemed necessary for the satisfactory accomplishment of all overhaul or repair work. Acceptance or rejection criteria for wear, deterioration, dimensions, operating parameters, alignment, and other standards of acceptance shall be specified, with limits listed in the text and summarized in appendix C. When required, machinery run-in operations shall be specified. Each component or major breakout item shall be defined by a separate chapter. If appropriate for clarity, a section shall be dedicated to each subassembly. Sections shall be identified by a capitalized Roman numeral as specified in 5.1.4. Each section (subassembly) shall begin on the top of a right-hand page.

5.8.1 Disassembly, inspection, and repair procedures. Step-by-step procedures for the disassembly, repair, and inspection of the repairable items shall shall be developed. Detailed instructions shall be limited to areas where the sequence of operations is not obvious or where special techniques are required for access to a part requiring repair. Where the sequence of disassembly is obvious, it is sufficient to state: "disassemble by removing parts in accordance with the disassembly sequence numbers". Additional instructions, cautions, and warnings shall be specified in sequence within a disassembly instruction as required when not obvious. Instructions for tagging, labeling, and match marking of parts shall be included in this paragraph. Where examinations and inspections must be performed prior to or during disassembly, the TRS shall so specify. Step-by-step procedures shall be provided in any of the following cases:

(a) Accomplishment of the required action would not be obvious to a journeyman mechanic unfamiliar with the particular equipment covered by the TRS.
(b) A peculiar or special action is required.
(c) Safety related or precautionary procedures are required to prevent personnel injury or equipment damage.
(d) Required action involves an area with a known history of failure due to improper procedures.
(e) Due to the complexity of a particular procedure, it is necessary to maintain the continuity of the TRS.
(f) Verification of the TRS demonstrates the need for a step-by-step procedure to satisfactorily complete the required action.

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5.8.2 Use of illustrations. Illustrations shall be used to ensure clarity of descriptive text and procedural steps. Exploded views shall be used to the greatest degree possible. Parts shall be numbered to show the sequence of assembly or disassembly and shall contain part identification by name, quantity, and other references (see 4.13.7). Redundant drawings shall not be included. Where only a part of an existing drawing is required, only that section of the drawing shall be provided.

5.8.3 Mandatory replacement parts. When the requirements exist for the planned mandatory replacement of particular items, the TRS shall reference the mandatory overhaul replacement parts list (see 5.12.1). The list shall be in tabular format in appendix A and shall identify those parts which are specifically required to be replaced, regardless of their condition. Materials shall be available at the overhaul activity in order to accomplish the planned material replacement actions in accordance with the TRS instructions. Examples of mandatory replacement items may be parts such as seals, washers, vacuum tubes, air filters, and motor brushes.

5.8.4 Examination criteria. Criteria to determine the suitability of worn or degraded critical parts for continued use shall be developed and shall state measurable wear or deterioration limits which, if exceeded, require corrective action to maintain reliability and specified performance. When it is required that this data be recorded, the specific measurement shall not be listed in the text. Rather, the text shall refer to appendix C where the data shall be listed in tabular form. If data are not to be recorded in appendix C, the data shall be presented in the text. Data shall identify the part involved and measurable limits. Departures from known existing specified limits shall be highlighted with footnotes as specified in 4.13.9.3 (see figure 6) or 4.13.11.1, as applicable. The required corrective action to be taken shall be specified in the text.

5.8.5 Special cleaning methods. When applicable to the repairable items covered in the TRS, special processes and methods of cleaning and examination to be performed on each disassembled item shall be described. Special cleaning and inspection instructions shall contain the following:

(a) The pre-cleaning examination requirements and cleaning process to be used for each item requiring cleaning.
(b) Step-by-step procedures for accomplishing the process.
(c) Specific identification of cleaning materials to be used in the process by their commonly known name and specification number. These data may be provided in tabular format.
(d) Quantitative instructions for the variables associated with the cleaning process (for example: air pressure, moisture content, solvent temperature, soak time, ultrasonic frequency, drying time, and temperature).
(e) Procedures for use of any special tools, jigs, or fixtures required during cleaning.
(f) Post-cleaning examination, preservation, and handling instructions.
(g) Cautions or warnings to be observed to protect personnel and equipment.
(h) Other data as may be appropriate for the repairable item type.

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5.8.6 Inspection, data collection, and corrective actions. Procedures, special instructions, examination, and technical information (including supplementary illustrations and tabular lists) shall be determined to perform effective inspection and repair of the repairable items covered in the TRS. The actions to be performed shall ensure that the repairable items meet or exceed the specified performance requirements of the final acceptance test and satisfy all quality assurance inspection requirements. This portion of the TRS shall, as a minimum, include the requirements specified in 5.8.6.1 through 5.8.6.5.

5.8.6.1 Fault identification. Detailed step-by-step procedures for inspection and location of faulty parts, misalignment, or other causes of trouble shall be identified. Where possible, examination procedures shall specify examination of the most expensive parts first. If irrepairable damage is found or the cost to repair is excessive, the decision can be made early in the overhaul to discontinue the overhaul and acquire a new unit.

5.8.6.2 Corrective action. Detailed step-by-step procedures for corrective actions (including the repair and replacement of defective, damaged, or deteriorated parts) shall be identified or developed.

5.8.6.3 Inspection and verification. Inspection and verification points shall be noted in the TRS left-hand margin by "I" or "V", both in the text and appendix C. I and V points shall be determined by using the best engineering judgment and the I and V attributes list. Data to be collected shall be specified in the text. The requirement to sign all I and V points shall be included in the front matter of parts 1 and 2 of the TRS. Signature blocks shall be provided as appropriate in the ETRAR (see figure 6).

5.8.6.4 Alignment and adjustment procedures. Detailed procedures shall be identified or developed for sophisticated mechanical and electrical alignments and adjustments. Procedures shall include detail of methods used and special tools and equipment required to achieve the setting and positioning of all adjustments and controls, and mechanical, gearing, and linkage alignments within the repairable item which, if maladjusted, would cause faulty operation.

5.8.6.5 Supplementary technical data requirements. If necessary, each section shall contain, within the text, all necessary supplementary technical information, illustrations, and tabular lists to aid the overhaul personnel in performing the specified actions. Such data shall include special information, instructions, methods, processes, and materials peculiar to the overhaul operation. Where applicable data are contained elsewhere in the TRS, such data shall not be repeated in this portion of the TRS but only referenced and their use specified. (Obvious repair actions, such as soldering and use of hand tools, shall not be included except where these actions involve hazards to personnel or equipment.)

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5.8.7 Reassembly. Illustrations, reassembly instructions, and examination requirements for the reassembly of the repairable items shall be developed as required. Where the sequence of reassembly is obvious, it is sufficient to state: "Reassemble by installing parts in accordance with the reassembly sequence number". Additional instructions, cautions, and warnings shall be specified in sequence within a reassembly instruction as required, when not obvious. Figures prepared to support disassembly instructions may be repeated; however, sequence numbers shall be changed to reflect proper numerical order for reassembly. The use of any special tools, jigs, fixtures, or test equipment shall be specified. The instructions or procedures shall indicate the following:

(a) The use of corrosion-preventive compounds, paints, or other materials and the use of gaskets or sealing compound materials which are not shop stores shall be specified by nomenclature, trade name, and specification number, as applicable.
(b) Identification of all points requiring lubrication, the kind of lubricants required, and the method of application. Lubricants to be used shall be those specified in the NAVSEA Standard Lubricants List of MIL-HDBK-267. A national stock number shall be listed if available.
(c) Any special instructions for installing fasteners, safety wiring, cotter pins, and other locking devices.
(d) Special instructions for aligning, adjusting, and measuring tolerances, clearances, end play, backlash, and tolerances between gears and linkage alignment data.
(e) Identification of critical examinations and tests during reassembly and, if required, provisions for data collection.
(f) Reference to tables, appendices, or illustrations as applicable.
(g) Other data as may be appropriate for the repairable item type.

5.8.8 Acceptance test. Acceptance testing shall be specified for each individual component if appropriate and for the system as a whole as required by NAVSEA. This section shall specify that:

(a) After accomplishment of all repair or overhaul actions and when the repairable item is considered ready for final examination, the item shall be tested to verify satisfactory performance.
(b) The acceptance test shall be accomplished in accordance with the established performance requirements. Reference shall be made to pertinent data contained elsewhere in the TRS.
(c) Specified data collection requirements shall be included in the TRS examination, test and repair action record for the item.

5.9 Chapter 6 (or as applicable) - unit assembly. This chapter shall specify all criteria applicable to unit reassembly. The following items as applicable to the unit shall be covered:

(a) Major unit installation.
(b) Piping/interface installation/alignment.
(c) Coupling cold alignment.

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5.10 Chapter 7 (or as applicable) - shop unit test. This chapter shall be developed to facilitate work breakout for a unit that requires a unit shop test prior to shipboard installation. A NAVSEA-approved outline for each generic unit shall be used in determining the number of chapters, chapter titles, and component/unit test requirements as follows:

(a) Test stand requirements.
(b) Unit flush.
(c) Unit shop test requirements.

5.11 Chapter 8 (or as applicable) - shipboard installation and alignment. This chapter shall specify the following requirements:

(a) Unit installation.
(b) Insulation installation.
(c) Cold alignment recheck.
(d) Piping/interface installation/alignment.
(e) System check.
(f) Coupling hot alignment.
(g) Unit break-in.
(h) Final acceptance test.
(i) Interfacing system inspection.

5.12 Appendices. The TRS shall provide supplementary and reference data required in the inspection, parts replacement, testing, and final acceptance of the repairable items covered in the TRS. Each appendix shall contain a statement delineating its purpose and application.

5.12.1 Appendix A - mandatory overhaul replacement parts list. Appendix A (see figure 4) shall contain a comprehensive list of items which must be replaced, regardless of observed conditions (see 5.8.3). There shall be a separate sheet listing the mandatory replacement parts for each unit subassembly or major subassembly component. The parts to be designated for the mandatory overhaul replacement parts list shall be specified by the preparing activity and shall be approved by the acquisition activity utilizing the following considerations:

(a) Parts which may be damaged or destroyed during disassembly.
(b) Parts subject to accelerated wear.
(c) Parts judged to be more economical to replace than to inspect.
(d) Parts which have a history of high usage.

The requirements of an overhaul shall not be met without fulfilling all listed requirements. In the event a material ordering guide (MOG) exists for an item being covered in the TRS, the MOG may be used to develop appendix A.

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5.12.2 Appendix B - contingency material list. A contingency material list (see figure 5) shall be developed. It shall be a tabular listing of parts which may require replacement during the overhaul of the unit. When developed, a separate sheet shall be prepared for each unit subassembly or major subassembly component. The contingency material list is a guide for material that may be needed when the unit is overhauled. Any part listed in the contingency material list shall be addressed in the appropriate inspection and repair subsection of the TRS. Those parts whose costs approach or exceed the total cost of the unit shall not be listed. Parts which rarely require replacement and are not normally replacement parts for unit overhaul shall not be listed. A MOG, if it exists, may be used to develop appendix B.

5.12.3 Material condition/overhaul report (MC/OR). A material condition/overhaul report (see figure 7) shall be prepared for inclusion in each TRS. The MC/OR, when filled in, will serve as a certificate of satisfactory completion of the overhaul and shall certify that the item meets all specified acceptance criteria. It will also provide a vehicle to collect additional information from the line mechanic pertaining to conditions observed prior to and during the overhaul. Directions for recording data and distribution of the report shall be specified in the TRS text. This form may be tailored to obtain specific information on the particular equipment. As a minimum, MC/ORs shall consist of the following:

(a) Check-the-box and short fill-in questions shall be used to the greatest degree possible.
(b) A heading identifying the component shall be completed by the preparing activity.
(c) Specific questions pertaining to the as-found material condition of the item shall be prepared.
(d) Provision shall be made for the mechanic to supply informal comments and short remarks.
(e) Specific questions or statements pertaining to the as-overhauled state of the item shall be prepared.
(f) Provision shall be made for the mechanic to record and highlight any approved or unapproved deviations and out-of-specification data.
(g) A separate MC/OR shall be prepared for each subassembly and major subassembly component and shall be included in the appropriate data section in part 2.
(h) MC/ORs shall be identified in the table of contents, list of effective pages, and TRS text.

5.12.4 Appendix C (part 2) - examination, test, and repair action record (ETRAR). Minimum information to be included in the table (see figure 6) shall be the system test step number, the desired response and tolerance limits, signature, and date. Other material may vary according to the nature of the equipment or user requirements. Separate summary sheets shall be developed for each subassembly and major subassembly component, as previously specified in the TRS, and assigned an appropriate title. Each instruction to record data in the text of the TRS shall have an entry in the ETRAR. Where possible, a line drawing identifying the measurement to be taken shall be provided on the same page. Readings, specifications, or parameters that depart from original design characteristics shall be highlighted and footnoted with rationale. A space for signatures shall

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be provided with date at all data entries. The last page in appendix C for each major subassembly component and the last pages of the unit shop test and unit final acceptance test shall specify the minimum criteria and acceptable values to assure class B overhaul requirements in accordance with NAVSEA test criteria. No space for recording data on these pages shall be provided; however, an instruction shall be included stating that local shipyard test forms shall be attached and submitted along with the completed overhaul record (part 2 of the TRS).

5.12.5 Additional appendices (D and on). Any other special requirements may be covered with additional appendices. When additional appendices are required, the text in the body of the TRS shall so specify.

5.12.6 Enclosures. Locally approved test data collection forms or logs shall be appended to part 2 of the TRS, when completed, for data maintenance purposes.

5.13 Technical manual deficiency evaluation report (TMDER). The preparing activity shall include three copies of the TMDER form behind each TRS (see figure 8). The TMDER shall be used to provide feedback on corrections and suggested improvements to the methods and procedures specified by the TRS. The TMIN and title shall be preprinted on each form by the preparing activity.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this standard.
(b) Quality program requirements (see 4.12).

6.2 Change considerations. In considering a change or a revision, the following information would expedite a decision by the acquisition activity.

(a) Number of pages in the proposed change or revision and percentage of pages affected in existing TRS by all changes.
(b) If the new revision merely adds pages to a chapter or to the back of the TRS, affecting no pages in the existing TRS, it should be so specified.
(c) Time required to prepare the new revision or change.
(d) Cost of preparing the new revision or change.

6.3 Deviations and waivers. Minor deviations from this standard may be approved at the local level with customer approval. Technical specification deviation and waiver requests shall be approved by the acquisition activity. Deviations and waivers to format, style, and TRS preparation quality assurance aspects are not authorized without the express written approval of the acquisition activity in accordance with established procedures.

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6.4 Submittals and approvals. When called for in the procurement contract, the following shall be submitted for approval:

6.4.1 TRS outline/bookplan. A TRS outline/book plan shall be submitted to the acquisition activity for review and approval, prior to the development of the TRS manuscript. Any significant change to the outline book plan shall be submitted to the acquisition activity for approval.

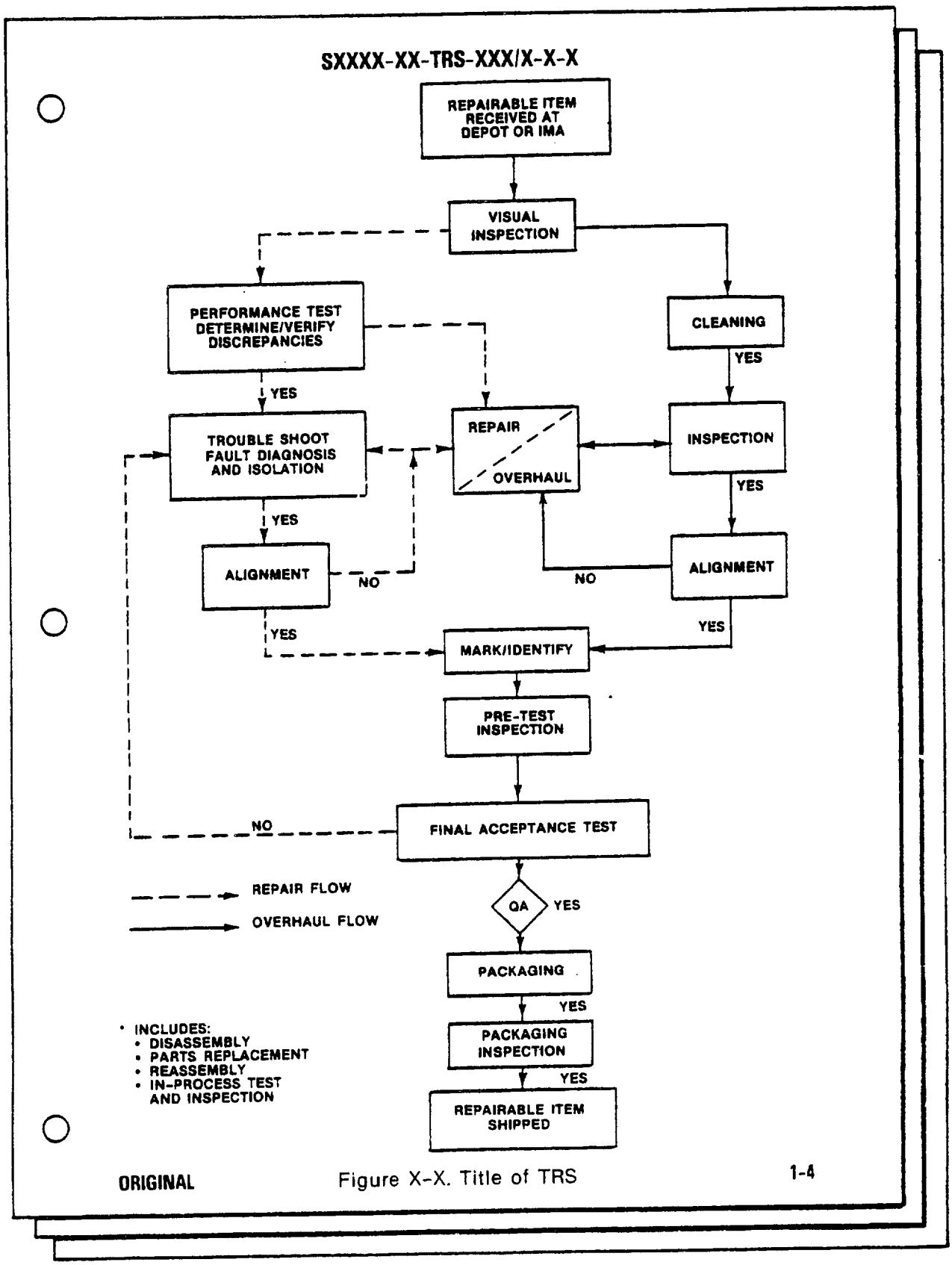
6.4.2 Manuscript TRS. The TRS manuscript shall be reviewed for technical content, adequacy and accuracy, and approved by the acquisition activity prior to submission of the final or preliminary TRS.

6.4.3 Preliminary TRS. The preliminary TRS shall review technical content, adequacy, accuracy, usability and ledgability requirements by the acquisition activity, prior to granting preliminary approval.

6.4.4 Final TRS. Reproducible copy and integral artwork shall be reviewed in accordance with the procedures for preliminary TRS, to ensure that verification comments have been incorporated, prior to the printing and distribution of the TRS. The acquisition activity shall grant final approval upon satisfactory completion of the TRS verification.

Preparing activity:
Navy - SH
(Project TMSS-N171)

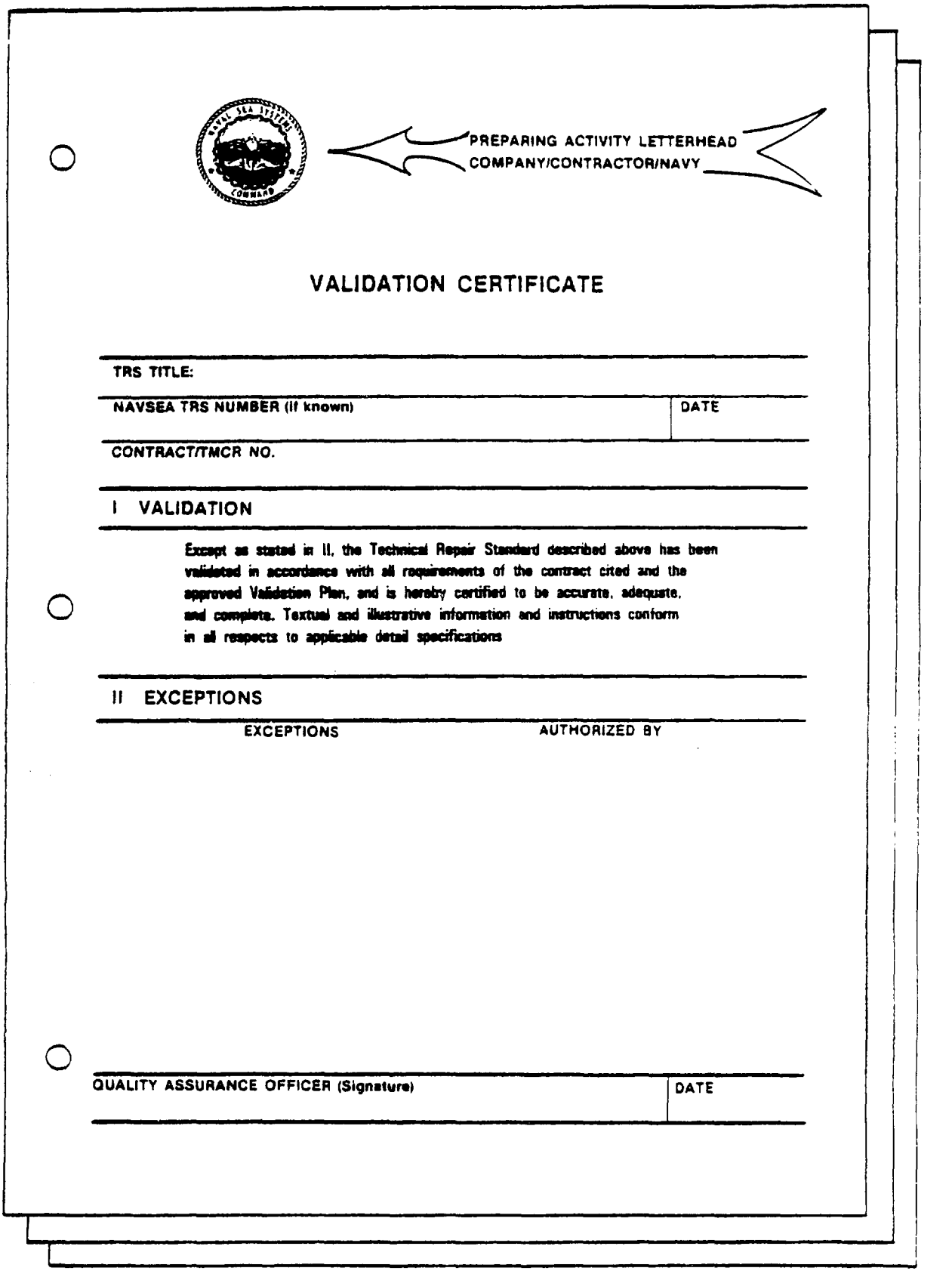
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SH 131928

FIGURE 1. Example of a repair process flow chart.

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SH 131929

FIGURE 2. Sample validation certificate.

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ENGINEERING JUDGMENT RECORD

TRS TITLE: _____

TMIN _____ APL(S) _____

TMCR/TMSR/BOA _____

PREPARING ACTIVITY _____

PARAGRAPH	JUSTIFICATION	REFERENCE SOURCE

GENERAL REMARKS

SH 131930

FIGURE 3. Sample engineering judgment record (EJR).

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ITEM _____

APPENDIX A

MANDATORY OVERHAUL REPLACEMENT PARTS

S9233-XX-TRS-XXX/XXXX-XXX

APL NO.	DESCRIPTION	NAVSEA OR MFG. DWG NO.	PC. NO.	MFG. PART NO.	QTY	NSN OR FSN

A-XX

SH 131931

FIGURE 4. Sample of a mandatory overhaul replacement parts list.

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ITEM _____

APPENDIX B

CONTINGENCY MATERIAL LIST

S9233-XX-TRS-XXX/XXXX-XXX

APL. NO.	DESCRIPTION	NAVSEA OR MFG. DWG. NO.	PC. NO.	MFG. PART NO.	QTY	NSN OR FSN

A-XX

SH 131932

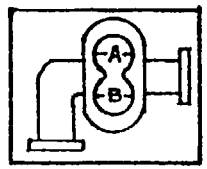
FIGURE 5. Sample of a contingency material list.

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Check the source to verify that this is the current version before use.

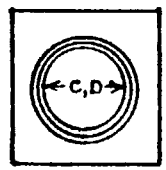
EXAMINATION, TEST AND REPAIR ACTION RECORD

SHIP NAME & HULL NO. _____ **TMIN** _____

ITEM/SUBASSEMBLY MN FD PUMP ATTACHED LUBE OIL PUMP **ITEM SERIAL NO.** _____



PART 6



PARTS 17&18



PARTS 19,20,21

	TRS PARA NO.	REF LETTER	DESCRIPTION	ACCEPTANCE CRITERIA	ACTUAL DATA INITIAL READING	ACTUAL DATA FINAL READING	CORRECTIVE ACTION TAKEN*	SIGNATURES	DATE
(I)	5.3.2.2	A	INSIDE DIAM. DRIVER GEAR BORE	2.400"-2.402"				I / V	
(I)		B	INSIDE DIAM. DRIVEN GEAR BORE	2.400"-2.402"				I / V	
	5.3.2.3	C	INSIDE DIAM. BRNG. BUSHING	0.908"-0.909"					
		D	INSIDE DIAM. BRNG. BUSHING	0.810"-0.811"					
(V)		E	INSIDE DIAM. BRNG. BUSHING	0.748"-0.749"					
(V)		F	INSIDE DIAM. BRNG. BUSHING	0.748"-0.749"					
(V)		G	INSIDE DIAM BRNG. BUSHING	0.748"-0.749"					

ORIGINAL

NOTE: *As per NAVSEA Ltr. 56Y21/BJW:jk Ser 907 of 16 Dec 1983
(USE BACK OF SHEET FOR REMARKS)

C-XX

SH 131933

FIGURE 6. Examination, test, and repair action record.

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S9233-XX-TRS-XXX/XXXX-XXX-XXX

MATERIAL CONDITION/OVERHAUL REPORT

SHIP: USS	HULL:

AS FOUND MATERIAL CONDITION (CHECK ONE)

☐ UNUSUALLY GOOD MAINTENANCE COULD HAVE BEEN DELAYED	☐ AS EXPECTED MAINTENANCE WAS NECESSARY	☐ UNUSUALLY BAD MAINTENANCE SHOULD HAVE BEEN DONE SOONER

REMARKS:

AS OVERHAULED CONDITION RECORD

	YES	NO
WAS ALL MANDATORY OVERHAUL REPLACEMENT MATERIAL (APPENDIX A) REPLACED?		
COULD MANDATORY OVERHAUL REPLACEMENT MATERIAL HAVE BEEN REUSED? (EXPLAIN IF YES)		
DID OVERHAUL ITEM MEET ALL THE ACCEPTANCE CRITERIA IN THE TRS?		

REMARKS: (EXPLAIN BLOCKS MARKED NO)

	YES	NO
WAS THERE AN APPROVED DEVIATION OF MATERIAL OR SPECIFIED PROCEDURES? (EXPLAIN IF YES)		

I CERTIFY, BASED ON PERSONAL KNOWLEDGE, THAT THE ABOVE ITEM HAS BEEN SATISFACTORILY OVERHAULED AND MEETS THE ACCEPTANCE CRITERIA PROVIDED HEREIN.

SHOP SUPERVISOR _____ DATE _____

ORIGINAL 3-XX

SH 131934

FIGURE 7. Material condition/overhaul report.

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NAVSEA (USER) TECHNICAL MANUAL DEFICIENCY/EVALUATION REPORT (TMDER)

NAVSEA 4160/1(3-82) (FRONT)
(Formerly NAVSEA 5600/2) (READ INSTRUCTIONS BEFORE FILLING IN FORM) (Use blank 8½" x 11" if more space required.)

INSTRUCTIONS
1. Use this report to indicate deficiencies, user remarks, and recommendations relating to the publication.
2. Blocks marked with * are to be filled in by Contractor before printing.
3. For unclassified TMDER's, fill in your return address in space provided on label printed on back of this form, fold on dotted line, staple, and mail to Commanding Officer, Naval Ship Weapon Systems Engineering Station,Code 5700, Port Hueneme, CA 93043. *Mark and mail classified TMDER's in accordance with OPNAVINST 5510.1E.*
4. For additional information, call Autovon 360-4805/5084 or Com'l. 805-982-4805/5084.

CLASSIFICATION OF THIS TMDER (Check Appropriate Block)
☐ UNCLASSIFIED ☐ CLASSIFIED Specify) _____

1. NAVSEA NO.*	2. VOL./PART*	3. TITLE*

4. REV. DATE (TM, CHG)	5. SYSTEM/EQUIPMENT	6. IDENTIFICATION/NOMENCLATURE (MK/MOD/AN)

7. USER'S EVALUATION OF MANUAL (Check Appropriate Block(s)) ☐ Excellent ☐ Good ☐ Fair ☐ Poor ☐ Complete ☐ Incomplete

8. GENERAL COMMENTS:

9. RECOMMENDED CHANGE(S) TO PUBLICATION

PAGE NO.	PARA-GRAPH	LINE NO.	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASONS

10. ORIGINATOR	11. RANK/RATE OR GRADE AND TITLE	12. DATE

13. WORK CENTER	14. TELEPHONE (AUTOVON/COM'L)

15. SHIP HULL NO. AND/OR STATION ADDRESS (DO NOT ABBREVIATE)

16. FOR NSDSA USE ONLY

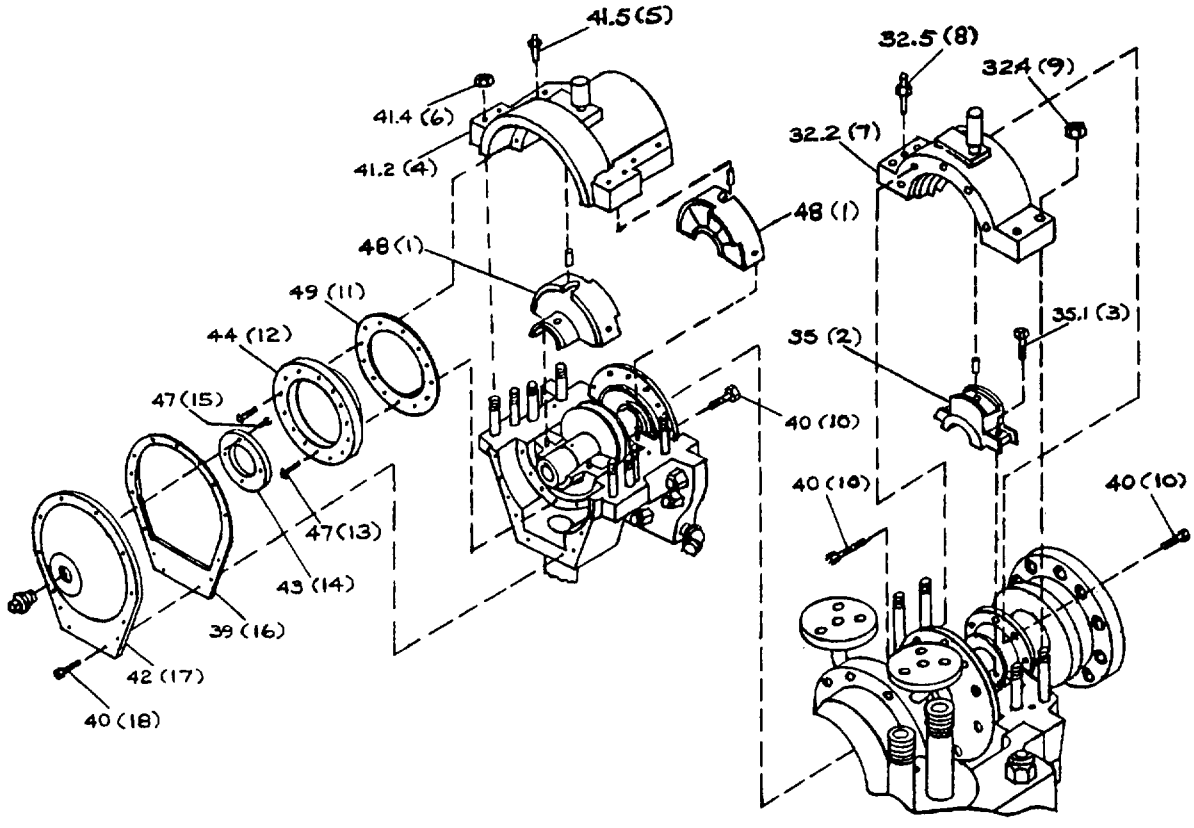
a. Control No.	b. COG ISEA	c. Date			d Priority	e Transmitted To
		Rec	Fwd	Due		

SH 131935

FIGURE 8. Technical manual deficiency evaluation report.

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S6225-A7-TRS-011/2552-086-600C



PARTS LIST

Part Number T.M.	Part Number SEQ	Part Name (QTY)	Part Number T.M.	Part Number SEQ	Part Name (QTY)
48	1	Thrust/Jrnl Brng Element	40	10	Hex. Head Bolt (28)
35	2	Line Brng Bushing Assy.	49	11	Laminated Shim
35.1	3	Shoulder Screw (2)	44	12	Thrust Brng. Retnr. Ring
41.2	4	Upper Half Housing	47	13	Socket Hd. Cap Screw (4)
41.5	5	Taper Pin (2)	43	14	Seal Ring Retainer
41.4	6	Hex. Nut (8)	47	15	Socket Hd. Cap Screw (18)
32.2	7	Upper Half Housing	39	16	Gasket
32.5	8	Screw Taper Pin	42	17	Thrust Brng. Hsg. Cover
32.4	9	Hex. Nut (4)	40	18	Hex. Head Bolt (13)

FIGURE 4-15. UPPER BEARING HALVES INSTALLATION

Original

4-21

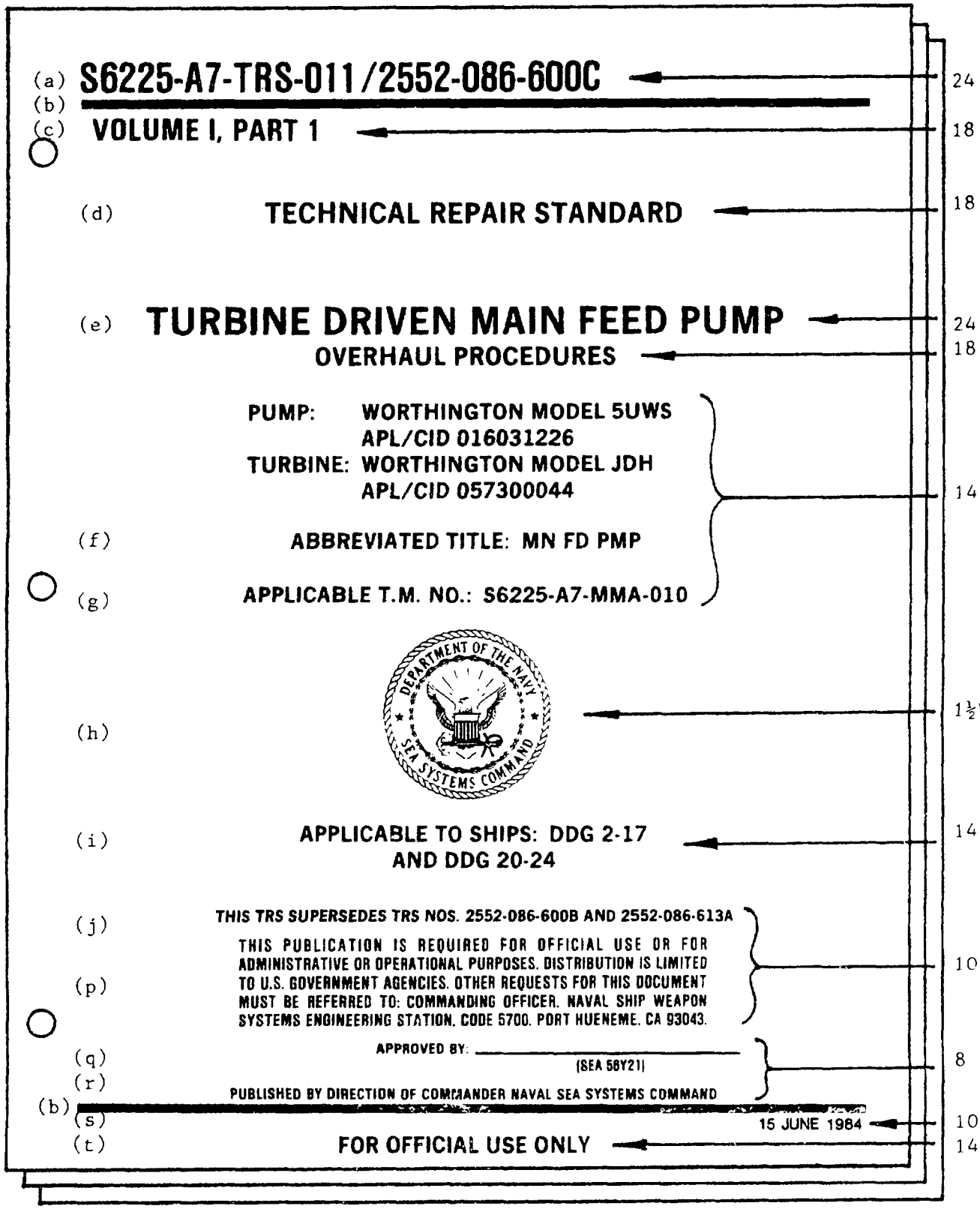
SH 131936

FIGURE 9. Typical illustration.

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(REDUCTION FOR ILLUSTRATIVE PURPOSES)

TYPE IN PT. SIZE



SH 131937

FIGURE 10. Sample TRS cover and title page.

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Notes to figure 10:

(a) The TMIN shall be requested from Naval Sea Data Support Activity (NSDSA) as specified in M0000-00-IDX-000/TMINS. The suffix display shall be requested from the appropriate PERA.
(b) A solid black line shall be placed below the publication number and above the date.
(c) Where more than one volume or part is required, the subject volume and part shall be noted below the line beneath the publication number.
(d) The words "TECHNICAL REPAIR STANDARD" shall appear in the upper center portion of the page.
(e) The nomenclature of the equipment, type, model, and allowance parts list numbers shall be positioned below the words "TECHNICAL REPAIR STANDARD". The classification of the equipment nomenclature shall be indicated (U), (C), (S), as specified in DoD Manual 5220.22-M, when the TRS itself is classified.
(f) An approved abbreviated TRS short title may be included in the location shown.
(g) If appropriate, an applicable technical manual identification number may be displayed in the title block.
(h) A 1-1/2-inch diameter acquisition activity seal shall appear centered on the page. Spacing between the necessary information shall be such as to result in an attractive, well-balanced title block.
(i) Applicable hull numbers shall be listed under the seal when appropriate.
(j) When a TRS supersedes a previous issue or other TRSs, a supersedure notice shall be in the space indicated.
(k) If the TRS contains proprietary information, the cover shall contain a suitable notice.
(l) TRSs containing classified information which may be liable to inadequate disclosure to foreign governments or foreign nationals shall bear a special handling notice.
(m) The espionage law notice shall be placed on the cover and title page for all classified manuals, including those bearing a "FORMERLY RESTRICTED DATA" marking, but not on those bearing a "RESTRICTED DATA" marking.
(n) National Security information notice is included when appropriate.
(o) Manuals classified "SECRET" (with or without "RESTRICTED DATA" or "FORMERLY RESTRICTED DATA" markings) shall show the total page count and copy number.
(p) The distribution statement shall be shown.
(q) The approval signature shall appear above the distribution statement on the title page only.
(r) A published by direction statement shall be included as shown.
(s) The publication date shall appear in the location shown below the horizontal line on the title page only.
(t) The words "FOR OFFICIAL USE ONLY" shall be included in the space shown.

FIGURE 10. Sample TRS cover and title page. - Continued

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TECHNICAL REPAIR STANDARD CHANGE NOTICE

ORIGINATOR NAME AND ADDRESS	TMIN: SXXXX-XX-TRS-XXX/X-X-X
	ITEM FSN/NSN XXXX-XX

DATE PREPARED	DATE ACCEPTED

ITEM(S) PART NUMBER(S)

ITEM NOMENCLATURE

PURPOSE:
This notice informs recipients that the standard identified by the number shown has been changed. The pages changed by this transaction being those furnished herewith and carrying the same date as this transaction. The pages, as dated and listed below in the summary of changed pages, combined with non-listed pages of the original issue of the latest approved revision, constitute the current version of this standard.

TRSCN NO.	PAGES CHANGED (INDICATE DELETIONS)	*S	*A	DATE
	PAGES CHANGED AND TRANSMITTED HEREWITH:			
	SUMMARY OF CHANGED PAGES:			

*S indicates supersedes earlier page. *A indicates added page.

ISEA/TECHNICAL APPROVAL	DATE
CTA/LCM APPROVAL	DATE

SH 131938

FIGURE 11. Sample of a technical repair standard change notice (TRSCN).

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PART 1

TITLE OF THE TECHNICAL REPAIR STANDARD

Front Matter
i Cover and Title Page
ii List of Effective Pages
iii Change Record
iv Foreword
v Table of Contents
vi List of Illustrations
vii List of Tables
viii Safety Summary

CHAPTER 1. - GENERAL INFORMATION

1-1 Scope
1-2 Point of Contact
1-3 Overhaul Conformance
1-4 References
1-5 Major Component Tabulation
1-6 Parts Replacement
1-7 Facilities, Test Equipment, and Personnel
1-8 User Comments Procedure

CHAPTER 2. - SHIPBOARD REMOVAL

2-1 Unit/Breakdown Combination
2-2 Unit Draining
2-3 Insulation Removal
2-4 Interference, Interface, and Piping Removal
2-5 Unit Installation Site (Foundation, Interface, and so forth.)
2-5.1 Preparation
2-5.2 Cleaning
2-5.3 Inspection
2-5.4 Repair
2-5.5 Blanking-off, Tagging, and so forth.

CHAPTER 3. - UNIT DISASSEMBLY

3-1 Major Component Boundary Identifications
3-2 Coupling Separation or/Interface and Piping Removal
3-3 Major Component Removal

CHAPTER 4. - MAIN FEED PUMP OVERHAUL PROCEDURES

4-1 Pump Disassembly
4-2 Parts Examination and Repair
4-2.1 Frame
4-2.2 Casing

SH 131939

FIGURE 12. Typical TRS subject matter arrangement.

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Check the source to verify that this is the current version before use.

4-2.3 Shaft
4-2.4 Impellers
4-2.5 Miscellaneous Parts
4-3 Pump Reassembly
4-4 Pump Shop Test

CHAPTER 5. - TURBINE OVERHAUL PROCEDURES

5-1 Turbine Unit Disassembly
5-1.1 Oil Piping Removal
5-1.2 Speed Limiting Governor Removal
5-1.3 Pneumatic Hydraulic Pressure Controller Removal
5-1.4 Hand Oil Pump Removal
5-1.5 Steam Chest Hand Valve Removal
5-1.6 Attached Oil Pump Removal
5-1.7 Motor Driven Oil Pump Removal
5-1.8 Governor Valve and Servomotor Removal
5-1.9 Turbine Removal
5-1.10 Oil Cooler/Strainer Assembly Removal
5-1.11 Low Suction Pressure Trip Removal
5-2 Turbine Disassembly, Inspection, Repair, and Reassembly
5-2.1 Turbine Disassembly
5-2.2 Turbine Parts Inspection
5-2.2.1 Turbine Frame
5-2.2.2 Turbine Casing and Exhaust Cover
5-2.3 Turbine Reassembly
5-3 Speed Limiting Governor
5-4 Pneumatic Hydraulic Pressure Controller
5-5 Attached Oil Pump
5-6 Motor Driven Oil Pump
5-7 Hand Oil Pump
5-8 Governor Valve and Servomotor
5-9 Steam Chest Hand Valve
5-10 Oil Cooler/Strainer Assembly
5-11 Low Suction Pressure Trip
5-12 Combined Exhaust and Relief Valve
5-13 Low Oil Pressure Alarm Switch
5-14 Lubricating Oil System Components
5-15 Base
5-16 Turbine Unit Reassembly
5.16.1 Low Suction Pressure Trip Installation
5.16.2 Oil Cooler/Strainer Installation
5.16.3 Motor Driven Oil Pump Installation
5.16.4 Hand Oil Pump Installation
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5.16.6 Steam Chest Hand Valve Installation
5.16.7 Pneumatic/Hydraulic Pressure Controller Installation
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5.16.9 Speed Limiting Governor Installation
5.16.10 Governor Valve and Servomotor Installation
5.16.11 Oil Piping Installation
5-17 Turbine Unit Shop Test

FIGURE 12. Typical TRS subject matter arrangement. - Continued

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CHAPTER 6. - UNIT ASSEMBLY

6-1 Major Component
6-2 Piping/Interface Installation/Alignment
6-3 Coupling Cold Alignment

CHAPTER 7. - UNIT SHOP TEST

7-1 Test Stand Requirements
7-2 Unit Flush
7-3 Unit Shop Test Requirements

CHAPTER 8. - SHIPBOARD INSTALLATION

8-1 Unit Installation
8-2 Cold Alignment Recheck
8-3 Piping/Interface Installation/Alignment
8-4 Insulation Installation
8-5 System Check
8-6 Coupling Hot Alignment
8-7 Unit Break-in
8-8 Final Acceptance Test
8-9 Interfacing System Inspection

Appendix A Mandatory Overhaul Replacement Parts List
Appendix B Contingency Material List

PART 2
EXAMINATION, TEST, AND REPAIR ACTION RECORD

Front Matter
i Cover and Title Page
ii List of Effective Pages
iii Change Record
iv Foreword
v Table of Contents

Appendix C Examination, Test and Repair Action Record
Shipboard Removal Data
Unit Disassembly Data
Pump Overhaul Data
Turbine Overhaul Data
Unit Assembly Data
Unit Shop Test Parameters
Shipboard Installation Data
Final Acceptance Test Parameters

Material Condition/Overhaul Report

FIGURE 12. Typical TRS subject matter arrangement. - Continued

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LIST OF EFFECTIVE PAGES

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Orig	S6225-A7-TRS-010/MN FD PMP	25 October 1978
Ch A	S6225-A7-TRS-01A/MN FD PMP	2 February 1980
Ch B	S6225-A7-TRS-01B/MN FD PMP	14 January 1982
Ch C	S6225-A7-TRS-01C/MN FD PMP	16 July 1982

Total number of pages in this publication is 158 consisting of the following:

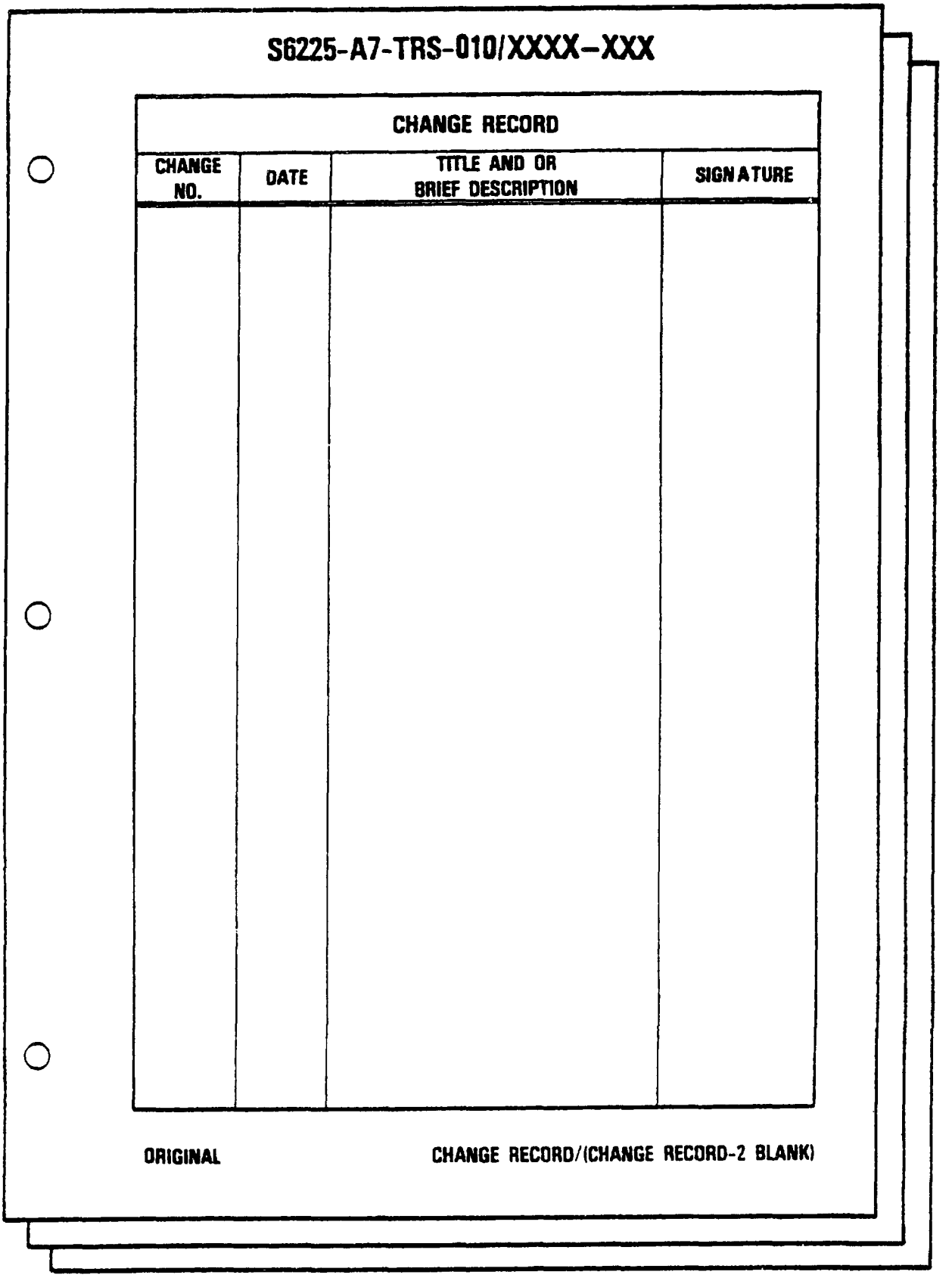
Page No.	*Change No.	Page No.	*Change No.
Title and A	C	3-1 through 3-47	A
Change Record	0	3-48 blank	0
Change Record-blank . . .	0	3-49 through 3-55	B
Foreword-1	B	3-56 blank	B
Foreword-2 blank	0	A-1 through A-15	0
i through xi	C	B-1 through B-4	B
1-0 through 1-17	A	C-1 through C-31	C
1-18 blank	0	C-32 blank	C
2-1 through 2-12	B	Index-1 through Index-4 . .	0

*Zero in this column indicates an original page.

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FIGURE 13. List of effective pages (LOEP).

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FIGURE 14. Sample change record.

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NOTE: The example above is hypothetical; the actual table of contents for a TRS is to be based upon the specific TRS content requirements for the repairable items covered by the TRS.

FIGURE 15. Sample table of contents.

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Note: When the list of illustrations and list of tables are brief, both may be shown on the same page.

FIGURE 16. Sample list of illustrations.

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Note: When the list of illustrations and list of tables are brief, both may be shown on the same page.

FIGURE 17. Sample list of tables.

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Safety Summary

The following warning notes appear in the text of this TRS and are repeated here for emphasis. Page numbers on which the warning notes appear are provided in parentheses.

Warning

Before removing or opening any fitting or part that can be subject to pressure (water or steam), ensure that double valve protection is provided with valves wired, preferably chained and locked closed. Follow the established ship's procedures for tagging-out valves. (Page 2-1)

Warning

Follow the established ship's procedure for tagging-out electrical equipment. High voltage, used in the operation of the motor driven oil pump and low oil pressure alarm contactor, can cause death on contact if personnel ignore safety precautions. (Page 2-1)

Warning

Asbestos is recognized as a major health hazard. Inhalation of invisible fibers of this material can lead to serious impairment of health in subsequent years. The insulation and lagging installed on the turbine and pump may contain asbestos material. (Pages 2-1 and 8-1)

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FIGURE 18. Sample safety summary.

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APPENDIX

DATA REQUIREMENTS

10. DATA

10.1 Data requirements. When this standard is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this standard is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.2	Technical manual status report	DI-M-2052	----
4.7.1	Record, engineering, judgment record, (EJR)	DI-L-2103	----
4.12	Manual technical; quality assurance, program plan	DI-M-2194	----
4.12	Manual technical; validation plan	DI-M-2195	----
4.12	Manual technical; validation certificate	DI-M-2196	----
4.12	Manual technical; verification incorporation certificate	DI-M-2201	----

(Data item descriptions related to this standard, and identified in this appendix, will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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10.2 The data requirements of 10.1 and any task in sections 4 or 5 of this standard required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this standard. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

10.3 Technical manual contract requirement/technical manual SEATASK requirement (TMCR/TMSR). Each contract or purchase order which contains the requirement for a TRS or TRS change pages should include an automated TMCR/TMSR, supplied by the Naval Sea Data Support Activity and approved by NAVSEA.

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